Agents Are Not Just Algorithms: The Tradeoffs of Decision-Time Reasoning in AI Trading

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Abstract

Agentic AI systems built on large language models generate decisions through sequential computation at decision time. We study how this process affects trading performance in a real-time market simulator where agents select tenders and execute trades. We experimentally vary reasoning intensity and market speed for agents based on frontier models. Additional reasoning improves tender selection and conditional execution, but delays action, causing missed trading opportunities and costly execution failures. Performance improves when the agent focuses on tender selection and delegates calculations and execution to compiled routines. Comparisons with a neural-network trader trained for the task show that the LLM agent's profits are more resilient to reduced liquidity. Decision-time reasoning can therefore add value beyond task-specific training, but its benefits depend on market conditions and the time available to act.

Keywords: Agentic trading, artificial intelligence, large language models, algorithmic trading

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Financial institutions are investing in artificial intelligence at substantial rates, increasingly deploying it as a decision-making technology. In capital markets trading, major dealers have built AI systems for execution: JPMorgan's LOXM platform, for example, uses deep reinforcement learning to execute equity trades.[1] Financial markets have also long been the domain of algorithmic trading more broadly. This raises a basic question: what features distinguish AI trading from algorithmic trading?

A growing finance literature has begun to take seriously the proposition that AI-driven trading is not simply faster algorithmic trading. Dou, Goldstein, and Ji (2025a) formalize this distinction and show that reinforcement-learning traders can autonomously sustain collusive equilibria. Dou, Goldstein, and Ji (2025b) study fragility under AI planning, and Dou et al. (2025) theoretically compare AI-powered investors with human investors. As this literature emphasizes, AI trading can differ from traditional algorithmic trading in economically meaningful ways. AI in financial markets has continued to evolve, however, and a newer class of *agentic* systems built on large language models (LLMs) has emerged (Lopez-Lira, 2025). These systems introduce a distinction within AI trading in how decisions are computed. A compiled policy applies a decision rule directly, whereas an LLM agent generates its response through a sequence of token evaluations whose length varies across decisions and with reasoning intensity. Both can have parameters fixed during trading; the relevant distinction is the computation required between observing the market and acting.

This paper studies the tradeoffs of decision-time reasoning in agentic AI trading. We show that additional reasoning can improve the quality of trading decisions, but it also consumes time as the market evolves. In real-time markets, performance depends on which parts of the task are left to decision-time reasoning and which are compiled into rules, calculations, or execution routines before deployment. Compiled algorithms, whether rule-based or learned through reinforcement learning, represent the fully compiled policy endpoint. To be clear, our question is not whether agents or algorithms dominate, but how and where decision-time reasoning is valuable.

We test these tradeoffs in a simulated market environment in which the trader faces a two-part task: deciding which client orders to accept, and executing the resulting trades.

[1] See `https://www.ft.com/content/16b8ffb6-7161-11e7-aca6-c6bd07df1a3c`.

In each session, the trader receives institutional requests to buy or sell blocks at quoted prices. The trader must decide whether to accept the tender and, if accepted, must unwind the resulting inventory in a real-time limit-order-book market. The task is useful because it separates two margins. Tender acceptance rewards judgment: the trader must evaluate quoted edge, displayed liquidity, remaining time, inventory, and penalties. Execution rewards speed: delay can cause a tender to expire before acceptance, and once inventory is created, delay exposes the trader to book movements and end-of-session close-out costs. The environment focuses on liquidity provision and execution, making both reasoning and latency economically meaningful.

We connect frontier language models to the simulated market through an interface that converts market states into model observations and model responses into actions. The prompt specifies the trading role and constraints, but does not prescribe an execution rule. The model chooses whether to accept or decline tenders, submit or cancel orders, or wait. We refer to this as the agent-in-control architecture because the model controls both tender selection and execution. Our baseline analysis uses `gpt-5.4` and varies the additional deliberation requested from the model across off, low, and medium settings. Even with reasoning off, the model generates its response sequentially from the current market observation. The simulator is the Rotman Interactive Trader (RIT; see Mak and McCurdy, 2019), which allows us to vary the wall-clock duration of a simulation tick. We use three market speeds—slow, normal, and fast—so that the same response time consumes different amounts of simulated market time. The design therefore varies both the requested deliberation and its market-time cost.

Figure 1 shows average net profits across reasoning settings and market speeds. Two patterns emerge. First, more reasoning does not consistently improve performance: the low-reasoning agent earns the highest average profits in two of the three speed treatments. Second, the return to additional reasoning also depends on market speed. In the fast market, the medium-reasoning agent earns negative average net profits because of fines for speculative positions and inventory remaining at session end. In the slow market, the same agent earns the highest average profits. These patterns illustrate how the balance between better decisions and delayed action changes with market speed.

Market speed also affects the downside risk associated with reasoning. In the fast market, medium reasoning (the highest level we consider) produces a heavier left tail of net profits than low reasoning, whereas in the slow market, the reasoning-off agent has the heavier left tail. Holding the reasoning setting fixed, slower markets tend to reduce the severity of losses. The cost of deliberation therefore appears in both average profits and downside risk.

Turning toward the mechanism, we first show that reasoning improves the agent's use of economic information at the decision point. Reasoning makes accept decisions more sensitive to current order-book conditions: more deliberative agents are more responsive to the spread between the tender price and what the agent could realize by walking the book, and they are less likely to accept tenders priced unfavorably to that spread. Conditional on completing the unwind, more deliberative agents also earn more profit from the trade. We refer to these improvements as a reasoning dividend.

The dividend, however, does not fully translate into net performance because reasoning consumes market time. We show that more deliberative agents attempt to accept more tenders that have expired during the agent's think time, and they leave more accepted positions stuck with residual inventory at session end. We refer to these costs as a deliberation tax. The tax is concentrated in faster markets, where each second of deliberation consumes more simulated market time. Thus, reasoning creates a dividend in selection and conditional execution, but incurs a tax in execution failures. The mechanism is therefore not better reasoning or slower response in isolation. Reasoning improves the agent's intended decisions and the quality of trades it does complete, but delay creates a wedge between intention and realized trading performance.

We also directly study the agent's recorded reasoning. More deliberative agents spend more of their reasoning explicitly evaluating whether an accepted position can be unwound, planning how aggressively to do so, and tracking remaining time and position limits. The content of the agent's recorded reasoning therefore aligns with what the dividend mechanism would predict: the agent considers task-relevant factors in its decision-making.

These findings suggest that the value of reasoning depends on where and when it is used. We first examine where reasoning belongs within the trading task by comparing the baseline agent-in-control architecture with three alternatives. The first provides pre-

computed variables, including book-based tender edge and projected inventory. The second combines LLM tender selection with algorithmic execution: the agent decides which tenders to accept, while a compiled routine unwinds the resulting positions. The final alternative is a fully compiled deterministic algorithm that handles both tender selection and execution without LLM reasoning.

Providing pre-computed decision support significantly improves performance over the baseline. Delegating execution to a compiled routine yields a further improvement: with the LLM agent responsible for tender selection, this architecture is competitive with the fully compiled benchmark. These results show the value of reserving decision-time reasoning for tender judgment while handling calculations and execution outside the reasoning loop.

Some adaptation can also occur before trading begins. We allow the agent to carry guidance from previous trading sessions into subsequent decisions through a written playbook revised between sessions. Performance improves on average, although the gains vary across configurations and costly execution failures persist in fast markets. Experience can therefore inform subsequent decisions without using trading time, but it does not remove the cost of reasoning when the agent acts.

Task-specific training offers another way to adapt before trading begins. We introduce an NN trader trained for the trading environment and find that increasing network size does not systematically improve performance or produce a detectable speed–performance tradeoff across the market speeds studied. We then impose the LLM agent's observed response delays on the NN trader and retrain its networks to account for delayed action. Retraining reduces the losses, but longer delays still lower profits. For the LLM agent, additional time also permits more reasoning, whose benefits outweigh the cost of delay in slow markets but not in fast markets. Training also leaves room for a benefit from decision-time judgment: when liquidity falls, the LLM agent's profits are more resilient than those of the NN trader. This resilience suggests that reasoning about current trading opportunities can add value beyond adaptation during training.

A natural concern is that the tradeoffs we identify reflect idiosyncratic features of `gpt-5.4` rather than a general property of agentic trading. We run the same task across additional frontier models from OpenAI and Anthropic. The varying effects of reasoning are robust,

but model behavior varies in ways that are themselves informative. For the Anthropic models we test, performance is highest with extended thinking turned off and declines as the reasoning budget rises. For smaller and less capable models in the OpenAI family, by contrast, additional reasoning can compensate for weaker baseline capability.[2]

Our main contribution is to show how the value of decision-time reasoning in AI trading depends on the time available to act. Additional reasoning can improve trading judgment, but the resulting delay can prevent those gains from translating into realized profits. The paper contributes to several literatures.

First, the paper contributes to a growing literature on AI in financial markets. Dou, Goldstein, and Ji (2025a), Dou, Goldstein, and Ji (2025b), and Dou et al. (2025) model AI traders as reinforcement-learning systems and study the equilibrium and welfare consequences of their behavior. Related work studies feedback loops and price dynamics when traders interact with learning algorithms (Banerjee and Szydlowski, 2025). Empirical work also uses deep learning and reinforcement learning as financial decision technologies, including deep sequence models for asset pricing and direct reinforcement-learning approaches to portfolio construction (Cong et al., 2021a,b). We extend this program to AI traders that reason at decision time, rather than executing a policy compiled in advance. A separate line of work uses large language models as predictive instruments: Lopez-Lira and Tang (2025) show that a general-purpose LLM can extract return-relevant information from text, and Lopez-Lira (2025) moves from passive inference to active trading by studying LLM agents in simulated markets. Relative to this line of work, we separate model capability from performance in a real-time market and study how the value of reasoning depends on the time available to act on it.[3]

Second, the paper connects to research on reasoning, test-time compute, and the economics of AI. In computer science, additional reasoning, tool use, and structured agent architectures can improve performance on many tasks (Wei et al., 2022; Schick et al., 2023;

[2]In the Online Appendix, we extend the design to a second task where the agent must decide how to route orders across two exchanges, and find similar reasoning patterns.

[3]The trading task we study also relates to a long literature on automated execution, dealer intermediation, and electronic liquidity provision (Hendershott, Jones, and Menkveld, 2011; Budish, Cramton, and Shim, 2015; Almgren and Chriss, 2000; O'Hara and Zhou, 2021; Hendershott, Livdan, and Schuerhoff, 2021). We use this task family as a laboratory for studying agentic reasoning.

Yao et al., 2023; Snell et al., 2025). Kang et al. (2025) identify related latency–quality trade-offs in high-frequency trading and gaming environments, and Hwang et al. (2025) highlight inference latency as a deployment constraint for LLM-based autonomous-driving architectures. In economics, AI is increasingly studied as a cognitive technology whose value depends on its interaction with judgment, task design, and organizational context (Agrawal, Gans, and Goldfarb, 2019, 2025; Korinek, 2025; Acemoglu, Kong, and Ozdaglar, 2026; Dell'Acqua et al., 2026). We add a dynamic market dimension: even if reasoning improves the decision conditional on a fixed state, the state may change before the decision is executed. The economic value of reasoning therefore depends not only on whether it improves judgment, but also on whether the environment leaves enough time to act on it.

Finally, our work points to a broader research agenda. AI agents differ from earlier automated trading systems not only in performance, but in the process by which they generate decisions. Recent finance work has examined this process from other angles: Bhagwat et al. (2026) use LLM personas to elicit economically meaningful investor disagreement and link it to abnormal trading volume, Chen et al. (2025) use interpretability tools to identify the concepts that drive LLM economic forecasts, and Bini et al. (2026) study systematic biases and corrections in LLM economic and financial choices. We study a complementary economic constraint: whether the agent's reasoning can be converted into trading performance when the market continues to move while the agent thinks. The economic content of AI decision-time reasoning and the conditions under which that reasoning produces value in financial applications are promising areas for future research.

1 Institutional Setting and Experimental Design

1.1 The Trading Task

We study a stylized electronic principal-liquidity-provision task. In modern institutional markets, intermediaries receive client requests electronically, decide whether to commit balance sheet, and, conditional on acceptance, must manage the resulting inventory through subsequent execution (O'Hara and Zhou, 2021; Hendershott, Livdan, and Schuerhoff, 2021).

The task captures features of modern electronic liquidity provision: rapid response to client flow, automated risk management, and execution across order-driven markets.[4]

The task has a natural two-stage structure. The trader must first decide whether to accept the client request, using the quoted edge, displayed depth, remaining time, and current inventory. Conditional on acceptance, the trader must then unwind the resulting position in the market by choosing order type, pacing, and, in some treatments, venue. This separation is valuable because it lets us distinguish two effects of reasoning. More reasoning can improve the quality of tender selection and inventory management. But reasoning also takes time, and during execution that delay is costly because the market continues to evolve while the agent thinks. The task therefore lets us observe both the *reasoning dividend*—the improvement in decision quality due to reasoning—and the *deliberation tax*—the cost of delay incurred by reasoning—in a setting where each is economically meaningful.

1.2 Platform and Simulated Market Environment

We implement the trading task in RIT (Rotman Interactive Trader), a pre-existing order-driven market simulator designed for real-time decision making in finance (Mak and McCurdy, 2019). RIT operates as a continuous double auction with price–time priority, supports limit and market orders, allows the market administrator to scale the market clock, and records participant actions at high frequency. These features make the platform well suited to our design because performance depends on decisions made under a real-time clock.

RIT also has the advantage of being a long-standing institutional simulator rather than an environment custom-built for this paper or tuned to the strengths of language-model agents. Rotman describes the RIT Market Simulator as a server–client centralized limit-order-book environment in which users submit orders, the server aggregates participant and AI order flow, and market-clearing prices are determined and recorded at millisecond frequencies.[5] The associated decision cases span buy-side and sell-side roles, including agency trading,

[4]For examples of firms emphasizing these activities, see `https://www.citadelsecurities.com/what-we-do/fixed-income-and-fx/`, `https://www.janestreet.com/what-we-do/overview/`, and `https://www.virtu.com/`.

[5]See the Rotman RIT overview: `https://www.rotman.utoronto.ca/faculty-and-research/education-labs/bmo-financial-group-finance-research-and-trading-lab/rit-market-simulator/overview--features/`.

liability trading, algorithmic trading, fixed income, commodities, derivatives, portfolio management, and risk management, and are designed to run through repeated scenarios with decision-support templates linked to the simulated markets through RTD and API connections.[6] Overall, our experiments use a documented, institutionally sponsored, longstanding market simulator with pre-existing cases, not a bespoke environment designed around the AI agents studied here.[7]

Our baseline environment uses the LT3 (Liability Trading 3) case. In LT3, the trader manages a liability book in two stocks, CRZY and TAME, over a five-minute, 300-tick trading session. During the session, the trader receives a sequence of institutional tender offers. Each tender is an offer to buy from or sell to the trader at quoted prices that vary in spread relative to prevailing market prices. The trader may accept or decline the tender. If the tender is accepted, it immediately creates an inventory position that must then be unwound in the market using market, limit, or marketable limit orders. The case implements fines for front-running, speculative trading unrelated to unwinding accepted tenders, and for carrying positions into the end of the trading session. It also imposes economically meaningful trading constraints, including net and gross position limits of 100,000 and 250,000 shares, maximum order sizes, transaction fees, and a tender decision window of approximately 30 seconds. Online Appendix Table A1 reports the full LT3 case parameters used in the baseline experiments.

Performance depends on decision quality in tender selection and execution skill rather than on forecasting market direction; the trader does not receive directional information about future prices. LT3 is a controlled environment that allows us to measure the effects of speed and reasoning on these performance dimensions. Importantly, we do not claim a literal mapping from simulator seconds to any particular real-world market.

[6] See the Rotman decision-case overview: `https://www.rotman.utoronto.ca/faculty-and-research/education-labs/bmo-financial-group-finance-research-and-trading-lab/rit-market-simulator/decision-cases/`.

[7] See `https://www.rotman.utoronto.ca/faculty-and-research/education-labs/bmo-financial-group-finance-research-and-trading-lab/rotman-international-trading-competition/sponsors/`.

1.3 AI Architecture and Reasoning Treatments

We connect frontier large language models to the market through a thin adapter layer. On each decision cycle, the adapter pulls the current market state from the RIT API, including the limit order book, inventory, active tenders, outstanding orders, and time remaining. It combines this market information with a system prompt that describes the trader's role, constraints, and available tools, and submits the resulting context to the model. The model's response is then translated back into market actions. The action set is intentionally narrow: the agent can accept or decline a tender, submit an order, or cancel an existing order. The adapter therefore connects the model to the simulator without hard-coding a trading policy. Figure 2 summarizes the two-stage tender-execution task and the agent architecture.

The prompt specifies the trading role rather than an execution algorithm. It instructs the agent to evaluate tenders, manage inventory, avoid speculation and front-running, and use available order types to unwind accepted positions. But it does not prescribe a fixed rule for when to accept, how aggressively to trade, or how to split execution across time. We structure the prompt in this way because our objective is to study an agent that reasons in real time, not an algorithm following pre-specified rules. Online Appendix Table A2 reports the baseline LT3 playbook prompt supplied to the agent alongside the market information that is provided to the agent.

We study frontier models from OpenAI and Anthropic under three reasoning settings: off, low, and medium. In all three settings, the agent receives the same market-state input and has the same action set. The treatment varies the requested amount of additional deliberation before the model returns an action.[8] The adapter presents the trading session to the model as a single evolving conversation rather than as a sequence of independent prompts. At each decision cycle, it appends the current market state to the same conversational thread, so the model can condition on the accumulated session history, including earlier tenders, prior actions, and their outcomes. Our baseline experimental analysis focuses on `gpt-5.4-2026-03-05`; provider and model portability are studied as robustness exercises.

[8]The agent uses decision-time cognition under all settings, including the off setting. The low and medium settings ask the model to deliberate more before responding. We do not include a high-reasoning treatment in the main specification because at high reasoning the associated latency dominates acceptance decisions.

1.4 Experimental Design and Protocol

Our main design crosses reasoning intensity with market speed. The reasoning treatments are off, low, and medium. The speed treatments are slow, normal, and fast. Because agentic systems reason in wall-clock time, realized think time is the natural object in our setting. The speed treatment determines how much simulated market time elapses during a given amount of wall-clock deliberation.

Specifically, let $\lambda_{env} \in \{0.5, 1.0, 1.5\}$ denote the number of simulation ticks per wall-clock second in each environment. One second of wall-clock time corresponds to 0.5, 1.0, and 1.5 simulation ticks in the slow, normal, and fast treatments, respectively. Equivalently, one simulation tick lasts 2.0, 1.0, and 0.67 seconds of wall-clock time. If an agent spends t_{think} seconds thinking, that delay consumes $\lambda_{env} t_{\text{think}}$ simulation ticks: $0.5\, t_{\text{think}}$ in the slow treatment, $1.0\, t_{\text{think}}$ in the normal treatment, and $1.5\, t_{\text{think}}$ in the fast treatment. The slow market therefore halves the simulation-time cost of deliberation relative to the normal market, while the fast market increases it by 50%. We disclose how much wall-clock time elapses per simulator tick to the agent in the system prompt to ensure it understands its environment.[9]

In total, the main speed experiment contains nine treatment cells: reasoning $\in$ {off, low, medium} crossed with speed $\in$ {slow, normal, fast}. In a normal-speed market with $\lambda_{env} = 1$, each run of the 300-tick LT3 case lasts for five minutes of wall-clock time. In the slow treatment with $\lambda_{env} = 0.5$, the same 300-tick run lasts for ten minutes; in the fast treatment with $\lambda_{env} = 1.5$, it lasts for three minutes and 20 seconds. These treatments should be interpreted as varying the amount of market evolution that occurs during a fixed amount of wall-clock deliberation within the simulator, rather than as imposing a literal calibration from simulator seconds to any particular real-world market. System prompts are identical across cells save for the disclosure of how much wall-clock time elapses per simulator tick.

This design separates a potential reasoning dividend from a deliberation tax. The dividend should appear most clearly in decisions that reward judgment, such as which tenders to accept and how to assess inventory and penalty risk. The tax should appear most clearly in

[9] The adapter also appends a fixed one-tick sleep between turns, pinned to the simulator's refresh cycle and held constant across treatments. Total wall-clock time per turn is therefore $1/\lambda_{env} + t_{\text{think}}$ seconds. In simulation time, this equals $1 + \lambda_{env} t_{\text{think}}$ ticks.

timely follow-through and position unwinding, because additional thinking leaves less time before a tender expires, the limit order book moves, or the session ends. The net effect on performance depends on the balance of these two margins at a given market speed.

The unit of observation is a session: one complete 300-tick run of a RIT case played by one trader. Across sessions, the structural case parameters are held fixed within an experimental cell, while the RIT environment generates randomized price paths and tender realizations for each run (i.e., no pairing). Each treatment cell contains 50 sessions. All language-model agents use the common observation-and-action protocol described above. We randomize cell order before execution.

The experiments are organized as treatment sweeps rather than as one fully crossed design. The primary design is the `gpt-5.4` reasoning-by-speed grid. The normal-speed `gpt-5.4` cells also serve as the baseline for the response-time analysis. Additional sweeps vary the architecture around the agent to study how moving tasks out of the decision-time reasoning loop shifts the speed–reasoning tradeoff, with deterministic algorithms as limiting points. We then run separate robustness sweeps across OpenAI model scale and Anthropic models. Because provider APIs implement reasoning controls differently, we interpret the provider comparison as a robustness exercise rather than as a matched reasoning-budget experiment.

2 Main Results

2.1 Summary Statistics

We begin with the reduced-form performance patterns for the baseline frontier agent, `gpt-5.4`. The main outcome is net liquidation value (NLV), defined as Trading P&L net of fines. Fines are economically important in this environment: they capture realistic costs from failing to respect trading mandates including front-running, speculative trades, and, importantly, carrying inventory into the end of the trading session. We also report Trading P&L separately because an agent can appear effective at extracting tender value while still destroying net performance through incomplete execution.

Table 1 reports cell-level means across the full reasoning-by-speed grid. Three facts stand out. First, the agent is economically active and can generate positive net performance. In the normal-speed market, mean NLV is \$8.3 thousand with reasoning off and rises to \$18.6 thousand under low reasoning. Low reasoning also performs consistently across speeds: mean NLV is \$22.9 thousand in the slow market, \$18.6 thousand in the normal market, and \$14.0 thousand in the fast market, and it has the highest "Sharpe ratio," which we use as a shorthand for mean scaled by standard deviation of performance, in each speed environment. The mean-performance ranking, however, depends on speed. In the slow market, medium reasoning earns the highest mean NLV, \$27.8 thousand, although its Sharpe ratio is slightly below that of low reasoning. This pattern is consistent with some real-time reasoning being useful in this task, but also suggests that the value of additional reasoning depends on how much market time the agent has to act.

Second, the relationship between reasoning and performance is not monotone. Medium reasoning does not dominate low reasoning. In the normal-speed market, medium reasoning produces higher Trading P&L than low reasoning, but also substantially higher fines, leaving mean NLV below the low-reasoning cell and slightly below the reasoning-off benchmark. In the fast market, the pattern is sharper: medium reasoning earns positive Trading P&L of \$14.5 thousand, but fines rise to \$33.0 thousand, producing mean NLV of −\$18.4 thousand. In the slow market, by contrast, medium reasoning earns \$34.5 thousand in Trading P&L and pays \$6.6 thousand in fines, producing the highest mean NLV in the grid. Thus the raw means already suggest that reasoning can improve some trading decisions, but that realized performance depends on whether the market gives the agent enough time to act on those decisions.

Third, the time cost of reasoning is large enough to be economically relevant. Average think time increases from roughly 2.7–3.2 seconds per turn with reasoning off to roughly 3.7–3.8 seconds under low reasoning and 5.7–8.4 seconds under medium reasoning, depending on the speed cell. Medium reasoning also accepts fewer tenders in the fast market than either off or low reasoning. These facts do not by themselves identify the cost of delay, but they show why latency is not a minor implementation detail. It is a treatment outcome that moves with reasoning intensity and can directly affect performance.

2.2 Reduced Form Evidence

Table 2 provides a reduced-form view of the speed-reasoning tradeoff in the normal-speed market, where the speed environment is held fixed. Columns (1), (3), and (5) regress NLV, Trading P&L, and fines on reasoning indicators alone. These specifications reproduce the non-monotone pattern in the raw means. Low reasoning raises NLV by about $10.3 thousand relative to reasoning off, while medium reasoning has essentially no raw NLV advantage. The decomposition shows why: low reasoning reduces fines, whereas medium reasoning raises Trading P&L but also raises fines.

Columns (2), (4), and (6) add log average think time per turn. We interpret these specifications cautiously because think time is a realized, post-treatment variable. The coefficients are not causal mediation estimates. They are nevertheless informative about the two margins that govern agent performance. Holding realized think time fixed, both low and medium reasoning are positively associated with NLV and Trading P&L. At the same time, longer think time is strongly negatively associated with NLV and Trading P&L and positively associated with fines. In economic terms, the table suggests that reasoning has a beneficial decision-quality component, but that the realized delay associated with reasoning is costly.

Figure 3 gives the visual counterpart to this regression. Each point is a normal-speed session, plotted by average think time and NLV. Within each reasoning level, sessions with longer average think time tend to have lower NLV. The medium-reasoning sessions also show greater dispersion and a thicker left tail, consistent with the idea that longer deliberation creates downside risk in a real-time execution task. The figure is descriptive, but it summarizes the empirical tradeoff: the relevant frontier is not model capability alone, but the joint effect of reasoning quality and the time required to produce it.

Taken together, the reduced-form evidence motivates the main experimental analysis. The raw performance facts show that low reasoning can improve performance, but that more reasoning is not automatically better. The response-time regressions show that the same treatment that improves decision quality also changes the amount of market time consumed before action. We therefore next use the experimental speed variation to test the

speed–reasoning frontier directly.

2.3 Experimental Evidence

The reduced-form evidence shows that reasoning intensity and realized response time move together. The summary statistics and Figure 1 illustrated the full reasoning-by-speed grid across the experimental variation. We now turn to formal tests across the grid.

Table 3 formalizes the speed and reasoning comparisons using NLV contrasts across cells. Panel A asks, within each reasoning level, how performance changes as the market slows relative to the fast-speed benchmark. The speed effects are weak when reasoning is off and under low reasoning. Under medium reasoning, the pattern is different. The joint speed test is statistically significant, and mean NLV is \$25.8 thousand higher in the normal market than in the fast market and \$46.3 thousand higher in the slow market than in the fast market. The pooled specification with reasoning fixed effects shows the same direction: the slow market raises NLV by \$15.7 thousand relative to the fast market.

Panel B turns the comparison around and asks, within each speed, how reasoning levels compare. We take low reasoning as the reference point because it is the strongest reliability benchmark in the main grid: it has the highest Sharpe ratio in every speed environment and the highest mean NLV in the normal and fast markets. The reasoning contrasts are speed-dependent. In the fast market, the joint reasoning test is statistically significant, but reasoning can hurt: medium reasoning lowers mean NLV by \$32.5 thousand relative to low reasoning, while reasoning off is \$3.4 thousand below low reasoning and statistically imprecise. In the slow market, the joint reasoning test is statistically significant for a different reason: medium reasoning is \$4.9 thousand above low reasoning, while reasoning off is \$20.3 thousand below low reasoning. In the pooled specification with speed fixed effects, both medium reasoning and reasoning off underperform low reasoning, by \$12.9 thousand and \$11.3 thousand, respectively.

Tables 4 and 5 show that speed and reasoning shift not only means, but the entire NLV distribution. In Table 4, slower markets compress the lower tail and performance range, especially under medium reasoning. The fast-medium cell has a first quartile of −\$50.0 thousand and a median of −\$8.9 thousand, even though its third quartile is positive.

Moving from fast to slow speed raises the first quartile by $56.7 thousand, the median by $44.8 thousand, and the third quartile by $18.8 thousand, while reducing the interquartile range by $37.9 thousand. Under low reasoning, moving from fast to slow raises the first quartile by $26.5 thousand and reduces the interquartile range by $29.2 thousand. The pooled quantile regressions show the same broad pattern: the slow market raises the first quartile and median relative to the fast market while compressing the interquartile range. Figure 4 reinforces this point visually.

Table 5 gives the companion view along the reasoning axis. In the fast market, moving from low to medium reasoning lowers the first quartile by $40.2 thousand and the median by $32.6 thousand relative to low reasoning, indicating that the fast-market cost of medium reasoning is concentrated in the lower part of the performance distribution. In the slow market, by contrast, medium reasoning is statistically similar to low reasoning across the reported quantiles, with higher point estimates for the median and third quartile as well as higher mean NLV. Reasoning off performs worse than low reasoning in the slow market, lowering the first quartile, median, and third quartile. The pooled quantile regressions summarize the same tradeoff: medium reasoning lowers the lower tail and median relative to low reasoning and increases the interquartile range, even though it can create upside when market time is abundant.

Overall, this evidence is consistent with a speed–reasoning frontier: the value of additional reasoning depends on whether the market gives the agent enough real time to act on the resulting decisions. Low reasoning performs best in the normal and fast markets. Medium reasoning is costly in the fast market, where its deliberation cost shows up in both mean performance and the lower tail of session outcomes. In the slow market, however, medium reasoning is no longer a net liability and has the highest mean NLV. We next examine the mechanism by separating the tender-selection margin from the execution and unwind margin.

2.4 Mechanism: Tender Selection vs. Execution Quality

The total performance results combine several decisions that occur at different points in the tender lifecycle. The agent must first observe a tender, decide whether the quoted price is attractive relative to the current book, successfully submit the accept decision before the

tender expires, and then unwind the resulting inventory before the end of the session. Here, we study each mechanism in detail.

2.4.1 Tender Selection and Fills

Table 6 studies tender selection and responsiveness. There are two key objects in this analysis. The first is the signal that the agent sends to the system to accept an offer, or "intent," given the proposed tender price and what it sees in the market. The second is the "book edge," or the spread between the weighted price of walking the tender through the current book and the tender price. Offers with a greater spread are more attractive, all else equal.

Panel A suggests the agent is conditioning on economically relevant information in the book, particularly when reasoning is on. It reports live-turn descriptive statistics, with one observation per agent turn while a tender is alive and before the agent has decided. The edge on turns with accept intent is higher than the edge on no-intent turns in every cell. In the normal market, for example, the mean edge on intent turns is \$0.1120 per share with reasoning off, \$0.1262 under low reasoning, and \$0.1290 under medium reasoning, while the corresponding no-intent edges are much lower, particularly when reasoning is on. Moreover, reasoning also tends to reduce the rate at which the agent accepts negative-edge tenders.

Panel B of Table 6 provides sharper regression-based evidence that reasoning increases the sensitivity of intent to edge. Current observed edge strongly predicts accept intent, with a greater slope when reasoning is on. Specifically, in the specification with reasoning-by-edge interactions, the edge slope is 0.683 under reasoning off, and the slope increases by 1.851 under low reasoning and by 1.827 under medium reasoning. The implied edge slopes under low and medium reasoning are therefore very similar, at roughly 2.53 and 2.51, and much steeper than the slope under reasoning off. Figure 5 visualizes the same result: the accept-intent schedule is much flatter with reasoning off and substantially steeper under low and medium reasoning.

This is consistent with a reasoning dividend: reasoning changes the sensitivity of the agent's actions to economically relevant information at the decision point. At the same time, Figure 5 makes clear that low and medium reasoning have higher baseline accept

propensity at zero edge as well as stronger edge sensitivity. Thus, while reasoning makes the agent more sensitive to edge, reasoning does not necessarily make every additional accept intent optimal.

Not every intent signal sent by the agent is filled by the system, for one of two reasons: either the tender has expired during the agent's own think time, or the implied post-fill position would breach the agent's position limit. In our sample, of the 71 failed intents, 52 occur because of tender expiration, and 19 occur because of the position limit.

Panels A and B show that greater intent-failure rates occur with medium reasoning. The percentage of failed intents for medium reasoning is 11% and 16% in normal and fast markets, compared to rates of 1–3% for off- and low-reasoning agents. The differences across reasoning modes largely disappear in the slow market, where all reasoning levels have a 1–2% intent-failure rate. Column (3) of Panel B provides regression-based evidence at the tender level. Low reasoning has essentially no effect on failed-intent probability, with a coefficient of -0.3 percentage points, while medium reasoning increases failed-intent probability by 7.1 percentage points. This pattern is consistent with an operational failure that can occur when real-time reasoning consumes a meaningful share of a fixed tender window.

2.4.2 Execution Quality and Overall Lifecycle

Table 7 studies post-fill execution quality. Panel A reports the post-fill unwind outcomes. Medium reasoning has the weakest unwind completion rates in the two faster environments: 78% in the normal market and 68% in the fast market, compared to a range of 83–89% for the off- and low-reasoning agents at those market speeds. In the slow market, completion rates are high across the board: 92 percent for reasoning off, 97 percent for low reasoning, and 91 percent for medium reasoning. This pattern matches the speed-frontier interpretation: when market time is less scarce, the same reasoning budget is less likely to leave the agent with residual inventory.[10]

Panel B columns (1)–(3) separate completion from conditional execution quality. Medium

[10] We count a tender as having a completed unwind if the agent's position crosses zero anytime after the tender is filled but before the session ends. Online Appendix Table A3 reports a fixed-window edge-capture check using an alternative measure that calculates ticker P&L over a deterministic 30-tick window after acceptance. This check addresses the concern that the completed-unwind measure could absorb variation in unwind duration. The alternative evidence is consistent with Table 7.

reasoning lowers the probability of completing the unwind by 9.8 percentage points, conditional on speed fixed effects, while low reasoning has essentially no completion effect. But among completed unwinds, both low and medium reasoning raise realized edge capture.[11] Controlling for book edge at the decision turn, low reasoning increases edge captured by \$0.024 per share and medium reasoning increases it by \$0.047 per share. Low and medium reasoning also raise the book edge at intent, by \$0.0126 and \$0.0155 per share, respectively. The table therefore shows both sides of the mechanism: reasoning improves selection and conditional execution quality, while medium reasoning creates more lifecycle attrition.

We also examine implementation shortfall (IS) to gauge the overall round-trip performance of the agent, defined following Perold (1988) and Hasbrouck (2007) as the per-share gap between paper-portfolio P&L and realized P&L, benchmarked against the bid-ask midpoint at the moment of acceptance.[12] IS provides a natural way to incorporate stuck unwinds into the analysis: the auto-liquidation cost on residual inventory enters as a separate component, distinct from the execution cost on traded shares. Panel A shows that off and low reasoning produce profitable round-trips, measured by negative IS, at all three speeds. Medium reasoning is also profitable in the slow market, with average IS of $-$\$0.090 per share, and is roughly break-even in the normal market. The fast/medium cell is the only condition where the trader incurs a clear net loss against the paper-portfolio benchmark, with average IS of \$0.158 per share. This reflects a stuck-unwind rate of 32 percent at fast/medium, where residual-inventory costs more than offset the edge captured on completed unwinds.

Panel B column (4) shows the corresponding regression on per-share IS. Medium reasoning increases shortfall by \$0.0647 per share on average, while the low-reasoning coefficient is negative and not statistically precise. The positive medium-reasoning coefficient is the

[11] Edge Captured is the realized per-share profit on a tender round-trip, computed as the change in the ticker's P&L between the acceptance turn and the unwind-end turn, divided by the tender quantity. The unwind-end turn is the first post-acceptance turn at which the ticker position has returned to its pre-acceptance level; if the agent fails to flatten before session end, the unwind is classified as stuck and excluded from Edge Captured. The measurement is gross of trading fees.

[12] Specifically, IS per share is $\frac{1}{Q}\big[-s(P_t-\pi_0)Q+s(\bar{p}_u-\pi_0)u+s(\pi_1-\pi_0)r+0.02u+1.00r\big]$ for a tender of side $s\in\{+1,-1\}$, quantity Q, fill price P_t, where π_0 is the acceptance midquote, π_1 is the post-stop settlement midquote, $\bar{p}_u$ is the unwind VWAP on u shares actually traded, $r=Q-u$ is the residual auto-liquidated at session end, and the last two terms are trading fees and the simulator's residual inventory penalty of \$1/share. Positive IS represents a drag on realized P&L relative to the paper portfolio. The trader is a liquidity provider, so negative IS values represent profitable round-trips.

net of two forces: medium reasoning improves execution on completed unwinds, but it also produces more stuck unwinds and residual-inventory penalties. Overall, the magnitude of the net shortfall effect depends on whether the simulation's pace gives the agent enough real-time budget to finish the unwind.[13]

To summarize, Figure 6 illustrates the lifecycle of a tender. Pooling across speeds, off and low reasoning produce 494 and 482 clean completed unwinds, respectively. Medium reasoning produces only 390 completed unwinds. It also produces more stuck unwinds, with 93 compared with 59 for off and 55 for low, and many more accept intents that expire during think time, with 45 compared with 1 for off and 6 for low. The figure shows why selection quality and total NLV can diverge. Medium reasoning can make more economically informed decisions, but many fewer tenders survive the full path from observation to accepted fill to clean unwind.

2.5 Reasoning Traces

Table 8 and Figures 7 and 8 provide complementary evidence on what the model does with additional reasoning. We classify each model's reasoning summary (or, in the case of reasoning-off, the text output returned by the model) into five categories using Claude Sonnet-4.6 as a judge/classifier.[14] For brevity, we focus on the normal market speed. The classifier evidence is not a performance mediation design, and we do not interpret it causally. It is useful as a check that the reasoning treatments change the content of the agent's deliberation in economically sensible ways.

Low and medium reasoning increase discussion of post-accept unwind feasibility by 24.4 and 29.4 percentage points, respectively. They also increase discussion of aggressive unwinding reasoning by 12.9 and 15.6 percentage points. Medium reasoning raises explicit time awareness by 9.9 percentage points and position-limit awareness by 2.0 percentage points. Figure 7 plots these coefficients, and Figure 8 shows representative traces in which the agent evaluates tenders, unwinds positions, and sometimes waits rather than immediately cross

[13] The pooled medium-reasoning coefficient on IS in column (4) masks substantial heterogeneity by speed. The fast/medium cell, where stuck-unwind rates are highest, drives the pooled positive coefficient.

[14] We use Claude to judge the reasoning traces of OpenAI models to guard against any potential bias from having a model judge the reasoning traces of models in the same family.

the spread after prices move against it.

Overall, the evidence to this point paints a clearer picture of the speed–reasoning frontier. Reasoning improves the agent's use of economic information: it makes accept decisions more sensitive to current edge, increases discussion of unwind feasibility, and is associated with higher edge capture conditional on successful completion. But lifecycle attrition is concentrated under medium reasoning: tenders expire during deliberation, accepted positions are more likely to remain stuck, and the lower tail of session performance deteriorates when market time is scarce. The mechanism is therefore not "better reasoning" or "slower response" alone, but rather a tradeoff between improved judgment and the market-time cost of turning that judgment into completed trades.

3 Agent Architecture

3.1 Moving Work Out of the Reasoning Loop

The mechanism results show that decision-time reasoning affects both the quality of the agent's intended decisions and the likelihood that those decisions are converted into completed trades. This raises an architecture question: which parts of the task should remain inside the agent's real-time reasoning loop, and which parts should be compiled into the surrounding system? We study this question by comparing four architectures in the LT3 normal-speed market.

The first architecture is the baseline agent studied above. The model observes the market state and chooses whether to accept tenders, submit orders, cancel orders, or wait. The second architecture provides decision support to the agent by adding pre-computed calculations to the agent's observation. This decision-support block reports book-based tender edge, projected inventory, limit-breach indicators, and flattening costs, but does not recommend an action.

The third architecture is agentic tender selection with algorithmic execution. The agent is invoked only when a tender is live and decides whether to accept each tender. It receives the

same decision support as in the second architecture.[15] Execution is then handled separately: a compiled unwind algorithm reduces any inventory back to zero by dispatching 5,000-share market orders each tick.[16] Thus, the agent retains responsibility for the tender decision, but it is not the continuous trading loop.

The fourth architecture is a fully compiled deterministic algorithm: it accepts tenders with a minimum 5-cent book edge net of fees and executes the same compiled unwind logic. This benchmark is distinct from the reasoning-off agent: in the off agent, the LLM still interprets the market state and chooses actions at decision time, while the algorithm applies a fixed tender rule and unwind routine. The algorithm is not intended as a horse race against the agent, but as the limiting case for what happens when both tender selection and execution are specified before the session begins.

Table 9 and Figure 9 report the results. Moving calculations out of the reasoning loop in the second architecture improves the agent substantially over the baseline. In the low-reasoning cell, mean NLV rises from \$18.6 thousand for the baseline agent to \$30.4 thousand with pre-computed calculations. Fines fall from \$7.0 thousand to \$1.5 thousand, the standard deviation of NLV falls from \$46.3 thousand to \$21.5 thousand, and the profitable-session rate rises from 74% to 96%. The gains are not limited to the low-reasoning cell: mean NLV also rises under reasoning-off and medium reasoning.

Agentic tender selection with algorithmic execution has an even larger performance boost over the baseline. With compiled unwind, the medium-reasoning agent earns mean NLV of \$48.5 thousand, pays zero fines on average, and has a Sharpe ratio of 2.38. The reasoning-off agent with compiled unwind earns a slightly lower mean NLV of \$44.5 thousand, is profitable in every session, and has the highest Sharpe ratio in this architecture, 2.85. These outcomes are competitive with a fully compiled algorithm that uses a 5-cent threshold to accept tenders, which earns mean NLV of \$44.5 thousand with a Sharpe ratio of 2.49. An algorithm that uses a 10-cent threshold to accept tenders earns a mean NLV of \$44.0 thousand with a Sharpe ratio of 1.77. Figure 9 shows the same pattern visually.[17]

[15] Online Appendix Tables A4 and A5 describe prompts for the agents in the second and third architectures.

[16] Bertsimas and Lo (1998) and Almgren and Chriss (2000) show that variants of this strategy can be cost-minimizing under certain assumptions.

[17] Within this architecture, the mean and Sharpe rankings differ: medium reasoning has the highest mean NLV, while reasoning off has the highest Sharpe ratio. Because Panel B and Figure 9 report mean-performance

Panel B of Table 9 formalizes the architecture comparisons using the cells with the highest mean NLV for each architecture, with the baseline low-reasoning agent as the omitted category. Decision support via pre-computed calculations with low-reasoning raises mean NLV by $11.8 thousand, although this estimate is imprecise, and reduces both the NLV interquartile range and fines. Agentic tender selection with algorithmic execution and medium-reasoning raises NLV by $29.9 thousand, increases Trading P&L by $23.0 thousand, and reduces fines by $6.9 thousand. The fully compiled algorithm is the compiled-policy endpoint and also improves substantially on the baseline agent, with mean-NLV gains that are comparable with the agentic-selection architecture.

These results sharpen the interpretation of the speed–reasoning frontier. The frontier is not fixed by the base model alone. It depends on what the agent is asked to compute and execute in real time. Decision support via pre-computed calculations reduces the burden of live evaluation. Agentic tender selection with algorithmic execution uses the agent where there is a potential reasoning dividend while mitigating the deliberation tax incurred by the child-order execution problem and LLM trading loop. With these supports in place, the agentic architecture improves significantly on the baseline agent and becomes competitive with the compiled-policy endpoint.

The implication of these results is not that agentic systems are better or worse than algorithms; it is that such systems should use decision-time reasoning where it belongs. Calculations and low-latency execution are natural candidates for compilation; agentic reasoning is most valuable where the remaining judgment is worth the delay.[18]

3.2 Reflection Between Sessions

The agent may also improve its decisions by learning from previous trading outcomes. We examine this form of adaptation in chains of three sessions, each with independently drawn price paths and tender schedules. After each of the first two sessions, a separate LLM reviews

architecture contrasts, we use the medium-reasoning cell as the representative Agent + DS + Algo Exec specification.

[18] Online Appendix Tables A6 and A7 report a separate prompt-ablation exercise in which we make the baseline prompt less directive without adding decision support; the ablation addresses the concern that the main results are driven by wording that mechanically pushes the agent toward urgent execution rather than by the information structure of the task.

the trading record and revises a written playbook supplied to the trader in the next session. Each chain starts without a playbook, and model parameters remain fixed throughout.[19]

Across the twelve configurations, mean NLV rises by $11.9 thousand between the first and third sessions, although the gains are uneven. The largest improvement is for `gpt-5.4-mini` with low reasoning, whose mean NLV rises by $71.1 thousand. Reflection does not consistently improve performance in every configuration, and substantial execution costs remain: the medium-reasoning `gpt-5.4` agent incurs mean fines of $20.1 thousand in the fast market even in its third session. These results suggest that experience can provide useful guidance before trading begins, while decisions made during trading remain subject to the cost of delay.[20]

3.3 Providers and Models

The main results use `gpt-5.4` as the baseline agent. A natural concern is that the speed–reasoning pattern reflects idiosyncratic features of this model, its provider-specific reasoning control, or the particular mapping from the reasoning label to realized test-time compute. We therefore examine whether the core patterns survive across model families. This exercise is not intended to rank providers. The reasoning controls are not identical objects across OpenAI and Anthropic, and the models differ in many dimensions besides reasoning budget. The purpose is narrower: to ask whether the central message is specific to one frontier model or whether it appears more generally.[21]

Table 10 reports descriptive means in the normal-speed market for three OpenAI models and two Anthropic models. Figure 10 plots the same evidence as connected means for NLV and think time. Three patterns emerge. First, performance varies substantially across

[19] We run 50 chains for each of twelve configurations: the nine reasoning–speed combinations for `gpt-5.4` and the three reasoning settings for `gpt-5.4-mini` at normal speed. The reviewing model is `gpt-5.4` with `xhigh` reasoning. Each playbook contains up to ten rules and replaces the previous version. Reflection takes place between sessions and consumes no trading time.

[20] Online Appendix Table A11 reports session means and within-chain changes.

[21] For OpenAI models, the off condition omits the reasoning field, while the low and medium conditions use bounded reasoning settings subject to the output-token limits in the runner. For Anthropic models, the off condition disables extended thinking while setting low output effort; the low and medium conditions set low or medium output effort and enable adaptive extended thinking without an explicit thinking-token cap. Thus the Anthropic labels are directionally aligned with the OpenAI labels but are not token-budget-equivalent. The providers also differ in API-level context handling and caching, so provider comparisons use realized think time and realized trading outcomes rather than treating the reasoning labels as identical primitives.

models. The baseline `gpt-5.4` agent earns positive NLV in all three reasoning cells, with the highest at low reasoning. The smaller OpenAI models are less reliable: `gpt-5.4-mini` is negative under off and low reasoning but improves under medium reasoning, while `gpt-5.4-nano` generates low average NLV and is especially weak under low reasoning. The Anthropic models perform strongly without extended thinking: Opus earns mean NLV of $28.4 thousand with reasoning off, and Sonnet earns $30.9 thousand.

Second, more reasoning is not uniformly valuable. For the baseline `gpt-5.4`, low reasoning improves NLV relative to reasoning off, but medium reasoning gives back the gain. For the Anthropic models, the pattern is even sharper: both Opus and Sonnet have their highest NLV with extended thinking off, and performance declines as the thinking budget increases. Medium reasoning is especially costly for Opus, where mean NLV falls to $4.7 thousand. In contrast, the smaller OpenAI models show some evidence that additional reasoning can compensate for weaker baseline capability, especially for `gpt-5.4-mini`. The value of reasoning therefore depends on the underlying model and provider implementation.

Third, the timing pattern is common across models. Figure 10 shows that higher reasoning levels generally increase average think time. The performance response, however, is not the same for every model: `gpt-5.4-mini` improves under medium reasoning despite the longer response time. The cells with very long response times nevertheless illustrate the downside risk of additional deliberation. Sonnet medium, for example, takes more than 16 seconds per turn on average and earns much lower NLV than Sonnet off or low; Opus medium also earns substantially lower NLV than Opus off or low. This pattern is consistent with the main interpretation: test-time reasoning can be useful, but its value depends on whether the additional cognition is worth the delay it creates in a real-time market.

These results support the external validity of the paper's main claim while also sharpening its scope.[22] The optimal reasoning budget is not universal. It depends on model capability, provider implementation, and the surrounding trading architecture. The robust feature is the non-monotonicity: increasing decision-time reasoning does not mechanically improve trading performance. Across providers and model scales, the economic object remains the

[22] In the Online Appendix, we show that the tradeoffs of decision-time reasoning are not specific to the LT3 task by considering a second task related to smart-order-routing.

same speed–reasoning frontier, but different models occupy different points on that frontier.[23]

4 A Neural-Network Trader

This section introduces an *NN trader* trained in a simulator of the LT3 case. It provides a benchmark for assessing the value of decision-time reasoning relative to a policy trained for the trading environment.

4.1 The NN Trader's Decision Problems

The NN trader uses separate networks for tender selection and inventory execution. The selection network decides which tenders to accept, while the execution network unwinds the resulting positions. Both networks are trained in an offline simulator of the LT3 case and evaluated in RIT. We begin with execution because the selection problem takes the execution policy as given. The Online Appendix provides further implementation details.

4.1.1 The Execution Network

The execution network chooses how quickly to liquidate a block of X shares over a *window* of H ticks, or until the session ends, whichever comes first. Let $t = 0, 1, \ldots, H$ denote elapsed ticks since the block arrives, and let x_t denote remaining inventory, with $x_0 = X$. At each tick $t < H$, the trader submits a fraction a_t of its remaining inventory as market orders. When these orders fill in full, inventory evolves according to $x_{t+1} = (1 - a_t)x_t$. The trader maximizes expected rewards,

$$\max_{\{a_t\}} \mathbb{E}\Big[\sum_{t=0}^{H} \gamma^t r_t\Big], \qquad r_t = -\frac{\mathrm{IS}_t}{X} \text{ for } t < H, \qquad r_H = -\Phi\frac{x_H}{X}. \tag{1}$$

where IS_t is the dollar implementation shortfall on shares traded at t, measured relative to the bid–ask midpoint when the block arrives and including the transaction fee of \$0.02 per share. The terminal penalty charges Φ per share of inventory remaining when the window

[23] Online Appendix Tables A8 and A9 provide regression versions of Table 2 separately for the OpenAI model scale comparison and the Anthropic family.

closes. Training uses $\gamma = 1$ and $\Phi = \$50$ per share. LT3 itself charges \$1 per share only on inventory held at session end. If fewer than H ticks remain in the session, the pacing window ends with the session. The execution policy conditions on time remaining, remaining inventory, a price input, and the previous liquidation fraction. The network's input vector is

$$s_t = \Big(\tau_t/H,\ x_t/X,\ \tilde{m}_t,\ a_{t-1}\Big), \tag{2}$$

where $\tau_t = H - t$ is the number of ticks remaining in the pacing window and $\tilde{m}_t$ is a rescaled last-traded price. The trader chooses from an action set $\mathcal{G}$ containing 21 liquidation fractions: zero and 20 positive fractions spaced geometrically between 2% and 100%. The action value $Q(s, a)$ is the expected payoff from choosing liquidation fraction a in state s and following the optimal policy thereafter. It satisfies the Bellman equation

$$Q(s, a) = \mathbb{E}\Big[r + \gamma \max_{a' \in \mathcal{G}} Q(s', a') \mid s, a\Big], \tag{3}$$

where r is the current reward and s' is the next state. At the last decision, the continuation value is the terminal payoff r_H in equation (1). The execution network approximates Q with a feedforward network $Q_w(s, a)$, where w denotes its parameters. It takes the four state inputs and returns one value for each of the 21 actions. The network has two hidden layers of 128 rectified-linear units. The trader chooses the liquidation fraction with the highest predicted value,

$$a_t = \arg\max_{a \in \mathcal{G}} Q_w(s_t, a). \tag{4}$$

We train the execution network from a random initialization using fitted value iteration, a batch form of Q-learning. The method uses simulated execution costs and subsequent market states to approximate the action values in equation (3).

4.1.2 The Selection Network

Tender selection balances the quoted price advantage against execution costs and the use of inventory capacity. For a tender of X shares, let e denote the per-share price advantage relative to the prevailing market price, and let $C(X, H)$ denote the expected cost of unwinding

the position over H ticks, measured against the same price. The tender's expected profit is

$$\pi = eX - C(X, H). \tag{5}$$

For an individual tender, positive expected profit requires an edge above the break-even threshold,

$$\theta = \frac{C(X, H)}{X}. \tag{6}$$

The inventory created by an accepted tender can prevent the trader from accepting another tender before the position is unwound. Any inventory remaining at session end also incurs a fine. We fit a state-dependent selection rule using session NLV to account for these costs At each session tick t when a tender is live, the selection network observes book edge, time remaining, gross inventory, and time since the previous acceptance. Its state is

$$z_t = \Big(\varepsilon_t,\ (T-t)/T,\ G_t/X,\ \min\{1, (t - t^{\text{acc}})/100\}\Big), \tag{7}$$

where $T = 300$ is the session length in ticks, G_t is gross inventory in shares, $X = 71{,}000$ is the inventory normalization, and t^{acc} is the tick of the most recent acceptance. The variable ε_t is book edge net of transaction fees, as defined in Section 2.4.[24] Unlike e, this measure already accounts for the estimated cost of walking the book. The selection network f_w has one hidden layer of eight units and 49 parameters. It accepts the tender when

$$f_w(z_t) \geq 0. \tag{8}$$

Otherwise, it waits. The network is evaluated once per tick while the tender remains live, allowing it to accept after market conditions change. A tender that is never accepted expires. We fit its parameters using mean NLV across simulated sessions, holding the execution policy fixed. The learned selection policy accepts a tender when its book edge net of transaction fee exceeds a state-dependent threshold. The Online Appendix reports the optimization procedure and training details.

[24] When depth is insufficient to absorb the full tender quantity, the remaining shares are priced at the deepest order book level.

4.2 Comparison with LLM agent

Table 11 compares tender selection by the NN trader and the LLM agent, with both using the same trained execution network. The LLM agent uses `gpt-5.4` with medium reasoning. We vary computer-trader order frequency across 120, 60, and 30 orders per simulation minute, holding the market clock fixed. These correspond to normal, half-frequency, and quarter-frequency markets. Each configuration is evaluated in 50 independently drawn RIT sessions.

The LLM agent's relative advantage becomes apparent when liquidity deteriorates. The two traders earn similar mean profits under normal liquidity, but at quarter frequency the LLM agent earns \$23.9 thousand compared with \$15.5 thousand for the NN trader. It remains profitable in 98% of sessions, compared with 58% for the NN trader. This performance is notable because the LLM was not trained specifically for the trading environment, whereas the NN trader was.

Both traders use the same execution policy and incur no inventory fines, so the comparison centres on tender selection and the timing of acceptance. The LLM agent's profits are more resilient to reduced liquidity than those of the NN trader. This resilience is consistent with a benefit of assessing trading opportunities at decision time when market conditions differ from the training environment. Such assessment consumes market time and carries a deliberation tax, but the smaller decline in profits suggests that the reasoning dividend retains value under deteriorating liquidity.

4.2.1 Network Size and Decision Delay

The previous comparison holds the NN trader's architecture fixed. We next vary network size, which allows greater flexibility in representing a decision rule at the cost of additional computation. We compare nine specifications of the selection network and nine of the execution network, varying one at a time. Each resulting trader is evaluated in 50 RIT sessions at each of the three market speeds. Larger networks do not systematically improve trading performance, and their performance does not vary systematically with market speed. Increasing network size therefore produces no detectable speed–performance tradeoff in our

setting.[25]

All these networks act with negligible delay at the market speeds studied. Varying network size therefore does not reproduce the LLM agent's variation in decision time. To examine the cost of delayed action and how far training can mitigate it, we impose delays drawn from the LLM agent's observed response-time distributions on the NN trader in each reasoning and market-speed treatment. The NN trader observes the market and chooses its actions, waits while the market evolves, and then submits those actions without revising them. When a tender is live, the median delay in the fast market is 5 ticks with reasoning off and 19 ticks with medium reasoning, relative to a tender window of 30 ticks. The corresponding medians are 3 and 13 ticks in the normal market and 2 and 7 ticks in the slow market. Each selection or execution decision still requires a single forward pass. The imposed wait therefore consumes market time without additional reasoning.

We compare the baseline networks, trained without delay (NN-unaware), with networks retrained under the imposed delays (NN-aware). Both the selection and execution networks are refitted separately for each treatment, retaining their architectures and state variables. Moving from the off-reasoning to the medium-reasoning delay distribution lowers mean NLV by \$34.3 thousand for the baseline NN trader and \$18.4 thousand for the retrained trader in the fast market. The corresponding declines are \$52.8 thousand and \$23.1 thousand in the normal market, and \$44.1 thousand and \$18.7 thousand in the slow market. All six declines are significant at the 1% level. The retrained NN trader remains profitable under the medium-reasoning delay, with mean NLV of \$13.7 thousand in the fast market, \$16.1 thousand in the normal market, and \$28.8 thousand in the slow market.

Lower inventory fines account for most of the retrained policy's advantage under the medium-reasoning delay. Its mean fines are \$4.2 thousand in the fast market, \$2.9 thousand in the normal market, and \$4.9 thousand in the slow market. The corresponding fines for the baseline policy are \$23.3 thousand, \$36.2 thousand, and \$30.2 thousand. In the fast and normal markets, longer delays also reduce the retrained trader's participation. Moving from the off-reasoning to the medium-reasoning delay lowers the share of observed tenders

[25] Online Appendix Table A14 reports performance across architectures and market speeds, together with joint tests of the architecture-by-speed interactions.

accepted from 62% to 27% in the fast market and from 56% to 29% in the normal market. In the slow market, however, acceptance rises from 61% to 71%. Across all three speeds, most of the retrained trader's off-to-medium decline in NLV reflects lower trading P&L. This decline in trading profits persists despite the different acceptance responses across markets.

For the LLM agent, the additional response time also permits additional reasoning. Moving from off to medium reasoning lowers its mean NLV by $29.1 thousand in the fast market, leaves it approximately unchanged in the normal market, and raises it by $25.2 thousand in the slow market. The retrained NN trader loses profits under longer delays at every speed, but its decline is not detectably larger in the fast market than in the slow market ($p = 0.97$). These results show that a cost of delay remains for the fitted NN policies even after training accounts for delayed action. The LLM agent can also earn a reasoning dividend by assessing the current trading opportunity at decision time. Together with the evidence in Section 2, this pattern is consistent with the reasoning dividend outweighing the deliberation tax in the slow market, but not in the fast market.[26]

5 Conclusion

This paper has examined AI traders that generate decisions through reasoning at the moment of trading. Reasoning produces a dividend in tender selection and the quality of trades the agent completes, but it also consumes market time, creating a tax through failed acceptance attempts and residual inventory. The balance depends on reasoning intensity, market speed, and the architecture of the surrounding system. Where cognition is placed matters: calculations and execution can be delegated to compiled routines, leaving the agent responsible for tender judgment. Comparisons with a neural-network trader trained for the task also reveal greater resilience of the LLM agent's profits to reduced liquidity. These results suggest that decision-time reasoning can add value beyond task-specific training, provided the market leaves sufficient time to act. The economic content of this reasoning, its interaction with system design, and its value under changing market conditions remain fruitful directions for future research.

[26] Online Appendix Table A15 reports the results across all reasoning and market-speed treatments.

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Figure 1: Decomposition of mean session NLV into Trading P&L and Fines across the reasoning × simulation-speed grid (gpt-5.4, $N = 450$; 50 sessions per cell). In each panel the upward green bar is mean Trading P&L, the downward red bar is mean Fines (plotted as −Fines), and the black tick marks mean NLV = Trading P&L − Fines. All panels share a common y-axis. NLV and Fines are annotated; Trading P&L is read off as the height of the green bar. The cell-level numerical values are reported in Table 1.

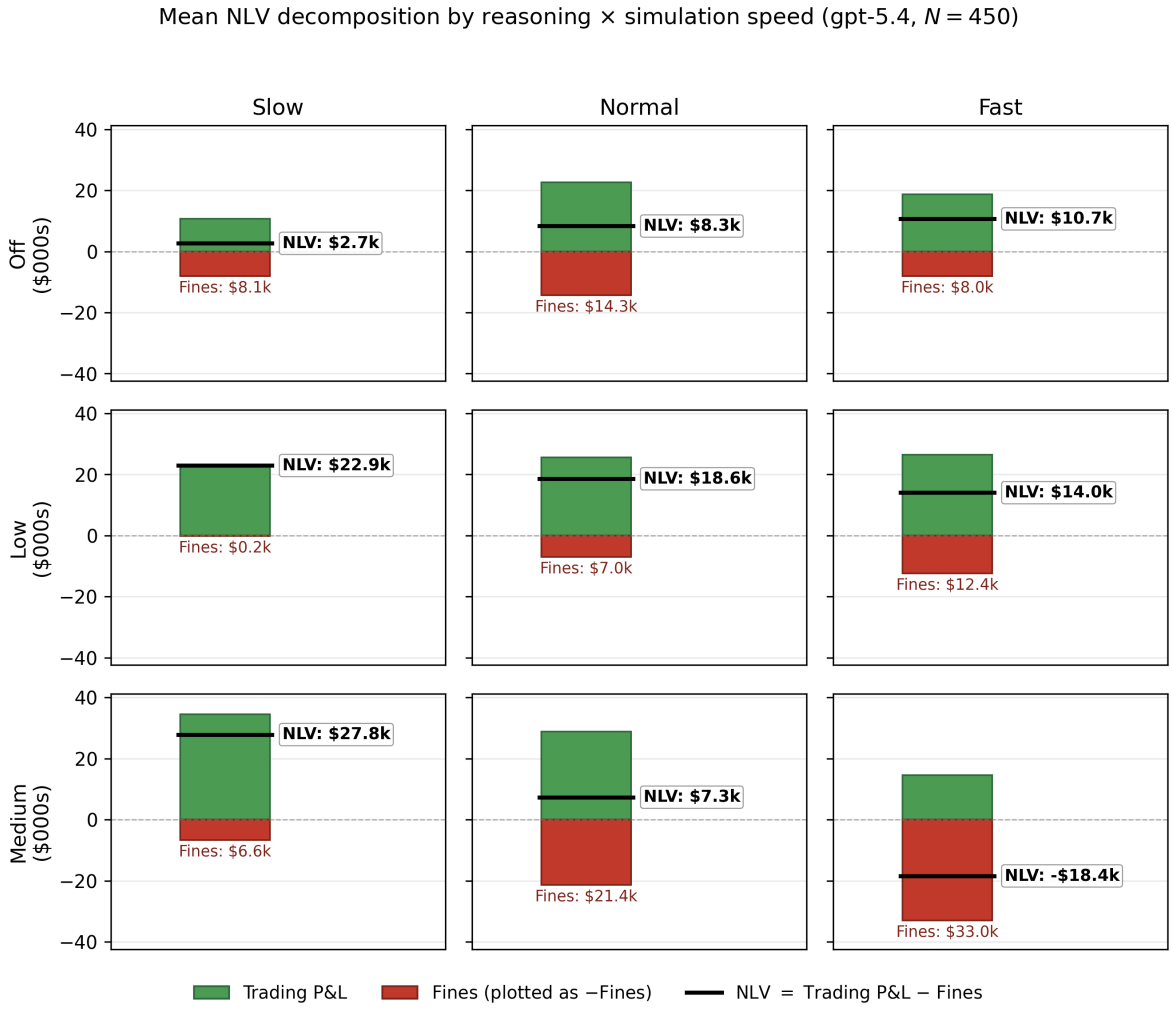
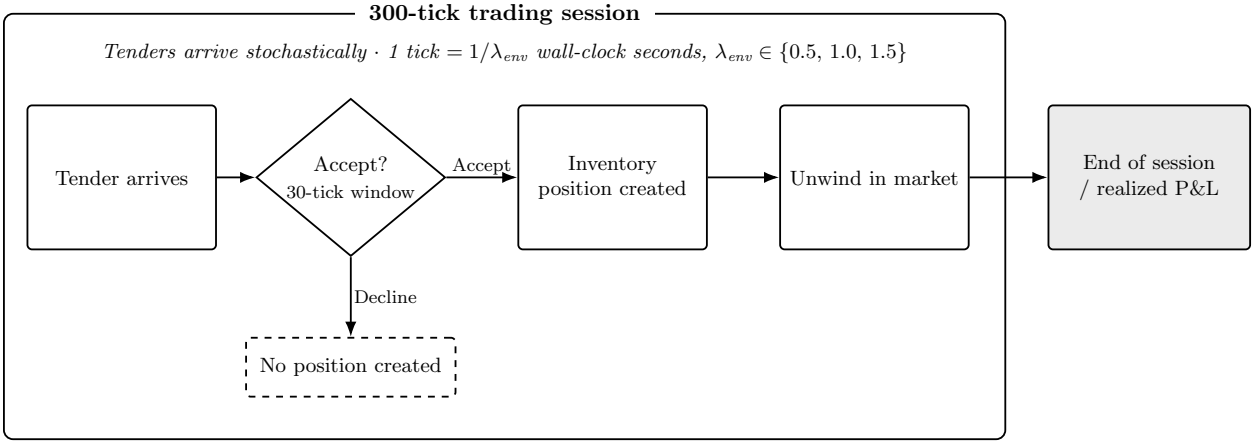


Figure 2: LT3 tender-execution task and agent architecture. Panel A summarizes the task. Each session lasts 300 ticks over two underlyings (CRZY, TAME), during which institutional tenders arrive stochastically. For each arriving tender the agent decides whether to accept; an accepted tender opens an inventory position that the agent then unwinds in the central limit order book. The session terminates at tick 300, at which point realized P&L is computed. Panel B sketches the agent architecture as an evolving conversation between the RIT simulator, a thin adapter layer (turn API, tool calls, response routing), and the frontier model (reasoning setting: off, low, or medium).

Panel A. LT3 Tender-Execution Task



Panel B. RIT-Agent Architecture

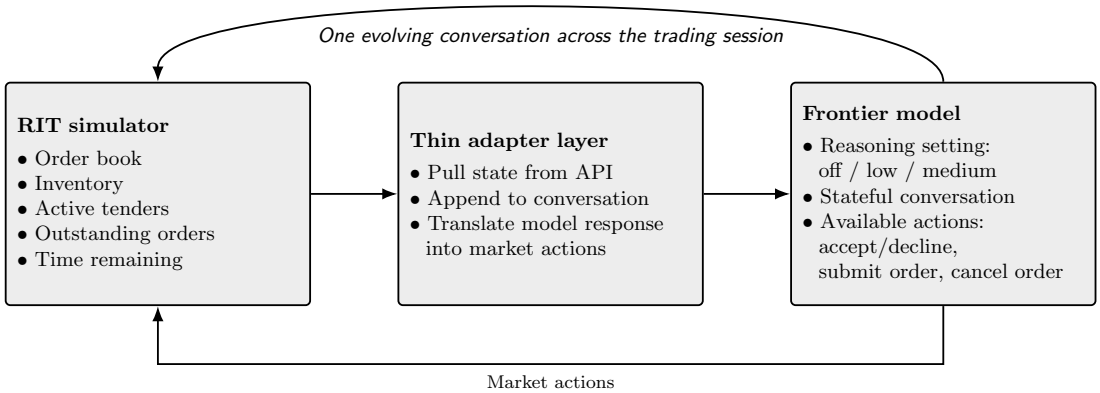


Figure 3: Session NLV vs. average think time per agent turn, gpt-5.4 in the normal-speed market ($N = 150$). Each point is one session, colored by reasoning level. Solid lines are within-reasoning OLS fits from $\text{NLV}_i = \alpha + \gamma_k \text{Reasoning}_i + \beta \ln(\text{ThinkTime}_i) + \varepsilon_i$. Filled markers are cell means. Horizontal axis is log-scaled; ln(ThinkTime) is a post-treatment variable.

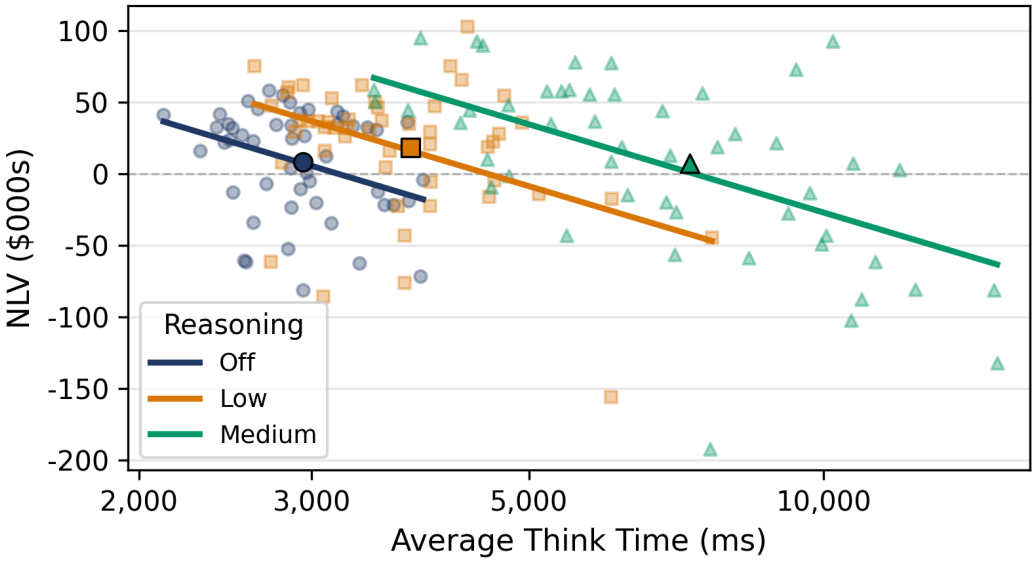


Figure 4: Per-cell distribution of session NLV ($000s) for the gpt-5.4 speed-pool experiment: 9 cells (rows = reasoning level, columns = simulation speed), 50 sessions per cell. Layout matches Figure 1. Boxes show the median (color line) and IQR (q_{25}–q_{75}); whiskers extend to the 10th and 90th percentiles. Diamonds mark cell means. The annotation in each cell reports the IQR ($q_{75} - q_{25}$) and the 80% spread ($P_{90} - P_{10}$) in $000s. The dashed line at zero NLV separates winning from losing sessions. All panels share a common y-axis.

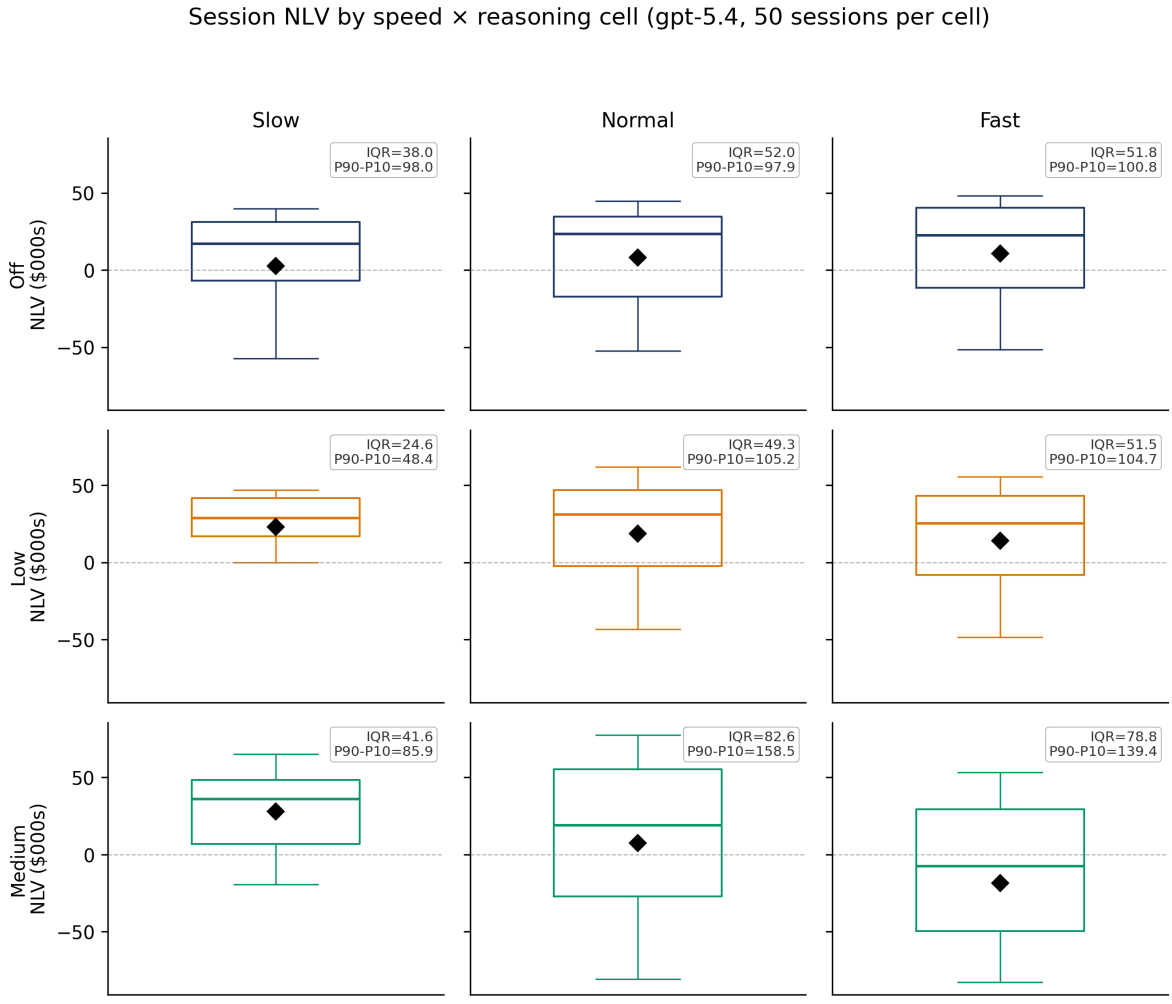


Figure 5: Predicted probability of accept intent as a function of the current observed book edge, by reasoning level (gpt-5.4; $N = 3{,}060$ pre-decision live turns across normal / fast / slow). Curves are pooled predictions from Table 6 Panel B column (2): a linear probability model with reasoning × book edge interactions and speed fixed effects, averaged over the three speeds. The dashed vertical line marks a zero-book-edge tender (break-even before fees); the dotted horizontal line marks $P = 0.5$. Tick marks at the bottom show the distribution of observed book edges (400-point subsample per level).

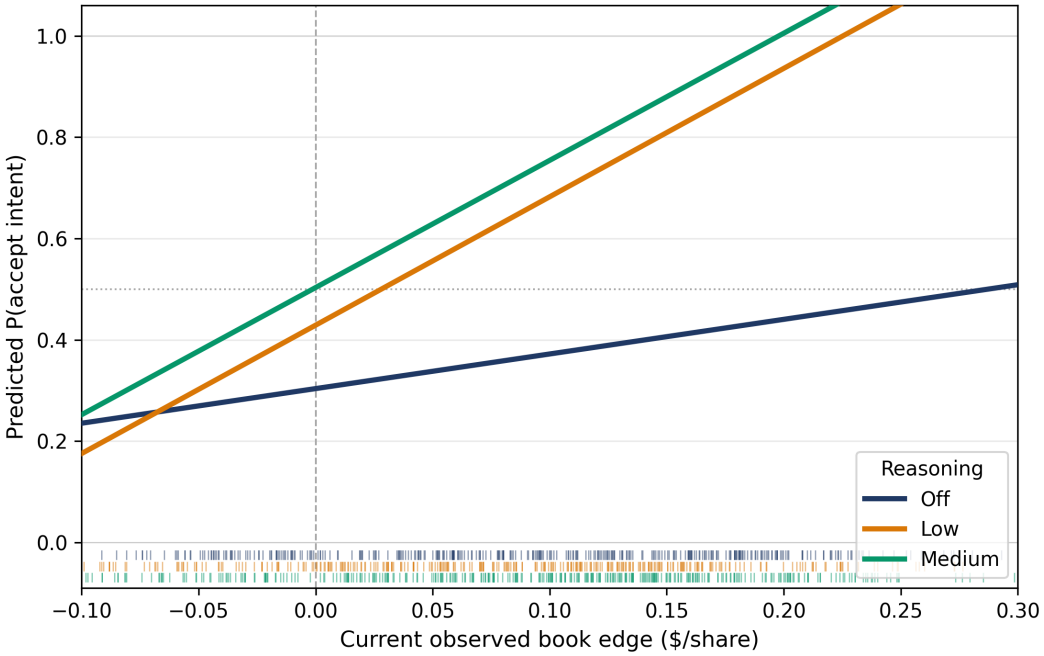


Figure 6: Tender lifecycle pooled across the three simulation speeds (gpt-5.4, 150 sessions per reasoning level). Each horizontal bar represents all tenders observed by the agent at that reasoning level (750/750/717 observed for off / low / medium). The medium count is 33 below the expected 750 because at fast simulation speed, Medium-reasoning think time was long enough that some tenders occurred and expired between the agent's polls. Segments — left to right — are the possible fates of a tender: the agent unwinds the accepted position cleanly (green); the accept fills but the agent ends the case with residual inventory (amber); the agent's accept intent fails to convert because the tender expires during the agent's own think time (red) or because the implied post-fill position would breach the 100k-share net limit (maroon); or the agent never issues an accept intent at all (grey).

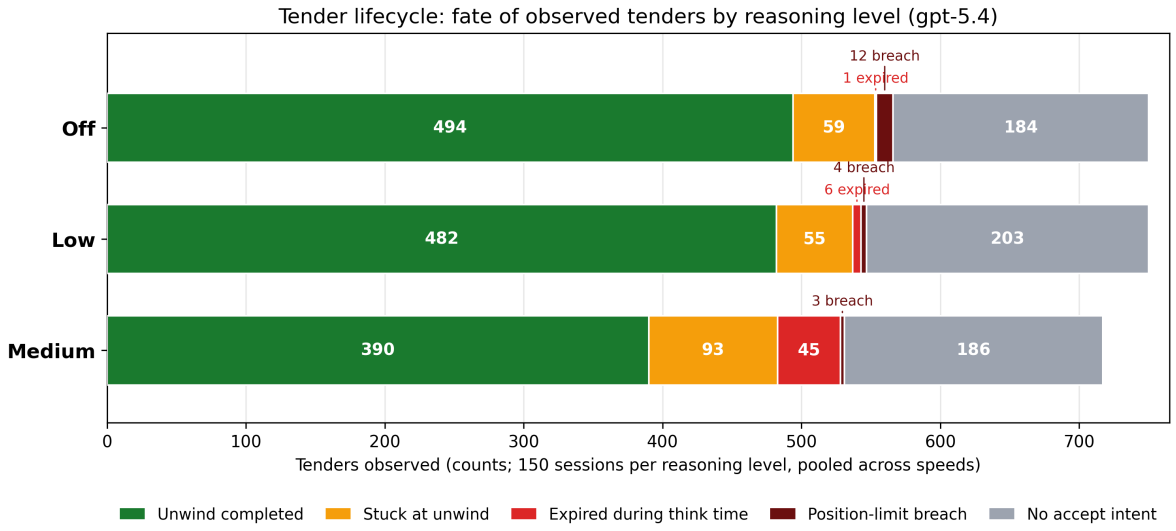


Figure 7: Coefficient plot for Panel B of Table 8: OLS coefficients from regressing each of five classifier dimensions on reasoning indicators (reference: off). Circle = Low, square = Medium; bars are 95% CIs, with bars that exceed the plotted range terminating in caret arrows. Sample: the 150 gpt-5.4 normal-speed sessions.

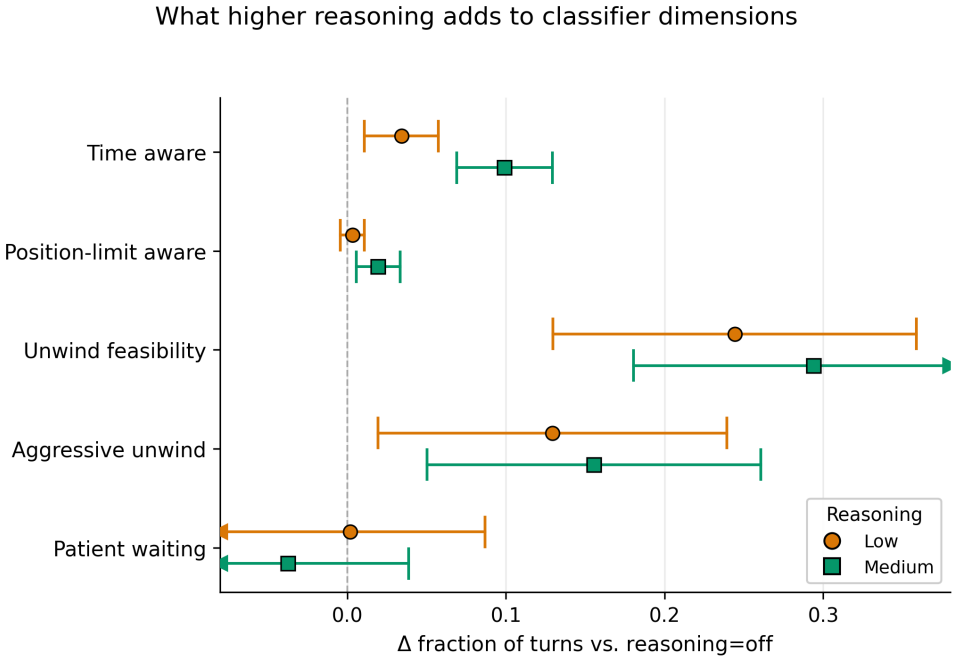


Figure 8: Sample reasoning traces from gpt-5.4 in the normal-speed market, one example per (reasoning level × situation) cell. Columns: (1) evaluating an incoming tender; (2) unwinding a just-accepted position into the book; (3) choosing to *wait* rather than cross the spread after prices move against the position. At reasoning = off the visible trace is the model's free-text `note` attached to its action; at reasoning = low / medium it is the OpenAI reasoning summary. Ellipses ("...") mark light trimming; section headers like **"Evaluating ..."** are emitted by the model, not added in post.

	Assessing a tender	**Unwinding a position**	**Waiting for a bounce (emergent)**
off	tender: buy 66k TAME @ 25.45 *"TAME tender is attractive: buy 66,000 at 25.45, while visible bids start 25.67 / 25.67 / 25.66 and remain above 25.45 for substantial depth. Even after $0.02/share unwind fees, expected edge is positive. Accepting now; will unwind on subsequent turn to avoid front-running fines."* **Action:** ACCEPT tender	position: −63,500 CRZY *"Hold. The accepted tender is profitable, and my resting CRZY buy orders at 10.08 are now top-of-book. No need to cross the spread yet; wait for fills and reassess next update."* **Action:** no orders, no cancels	−74,000 TAME, mark P&L −$31,820 *"Hold. The TAME tender is currently deeply unprofitable to unwind, so I should not chase the market higher. Keep the 25.80 bids resting and wait for mean reversion / liquidity to come back."* **Action:** no orders, no cancels
low	tender: sell 70k CRZY @ 10.46 ***"Calculating budget constraints.*** 605,310/59,100 ≈ 10.24. *I still need 10,900 at avg above 10.5 but under 10.44. Total budget for 70,000 at avg 10.44 = 730,800. After covering 605,310, remaining budget 125,490 — so up to 11.51 on the last 10,900."* **Action:** ACCEPT tender	position: −29,400 CRZY ***"Evaluating trade strategy.*** *Cover remaining 29,400 with asks totaling 26,300 at 10.45; fill 25,000 there, fully profitable. ... Risk if market rises — profits already dropped 24.5k → 16.5k. Covering 25k at 10.45 is an acceptable small loss to secure most of the position."* **Action:** BUY 25,000 CRZY ML @ 10.45	−69,000 TAME, ask > tender price ***"Analyzing market orders.*** *Market shifted against us; short at 24.82, best ask now 24.93 → loss if we cover. Waiting for a possible mean reversion seems reasonable, and our 24.81 bids might get filled if the price retreats. ... Hold our resting bids; see how the market moves."* **Action:** no orders, no cancels
med.	tender: sell 66k TAME @ 25.10 ***"Evaluating share tender profitability.*** *... want to cover at avg price less than $25.08 (after $0.02 fee).* ***Calculating cumulative share costs:*** 10,000@24.87 = 248,700*;* +600@24.87 = 263,622*;* +9,400@24.91 → 497,776 *for 20k ...* ***Calculating potential profits:*** *profit of about $8,201 ... acceptance of the tender is the next move!"* **Action:** ACCEPT tender	position: −68,000 TAME ***"Deciding trading strategy.*** *... marketable limit around 25.14–25.15 ... 7 × 10k at 25.14 might fill up to that price.* ***Calculating profitability:*** *full 68k through ≈ 25.22, average ≈ 25.05. Good profit cushion → buy aggressively, mindful of liquidity moving.* ***Planning simultaneous orders:*** *sending 7 orders at once ..."* **Action:** 7× BUY 10,000 TAME ML	−8,900 CRZY, resting bid @ 10.20 ***"Evaluating bid strategy.*** *... best bid is also 10.20, best ask is 10.23. Crossing at 10.23 is a loss after fees. Better to wait at 10.20 ... ensures a small profit if filled.* ***Deciding on bids:*** *since it's the top bid, there's a chance it could fill ... I don't need to make any tool calls right now; I'll just wait and see if it fills."* **Action:** no orders, no cancels

Figure 9: Four architectures applied to the LT3 case, 50 sessions per cell. Agent and Agent + DS use Low reasoning; Agent + DS + Algo Exec uses Medium reasoning; the Algorithm row uses the $0.05 tender-acceptance threshold. **Panel A**: per-session NLV distributions (box: 25th / 50th / 75th percentiles; whiskers: 1.5×IQR; black diamond: mean; jittered dots: individual sessions). **Panel B**: accounting decomposition of mean NLV into mean Trading P&L (green bar above zero) and mean Fines (red bar below zero); the black tick marks mean NLV = Trading P&L − Fines. Numerical values and pooled regressions are in Table 9.

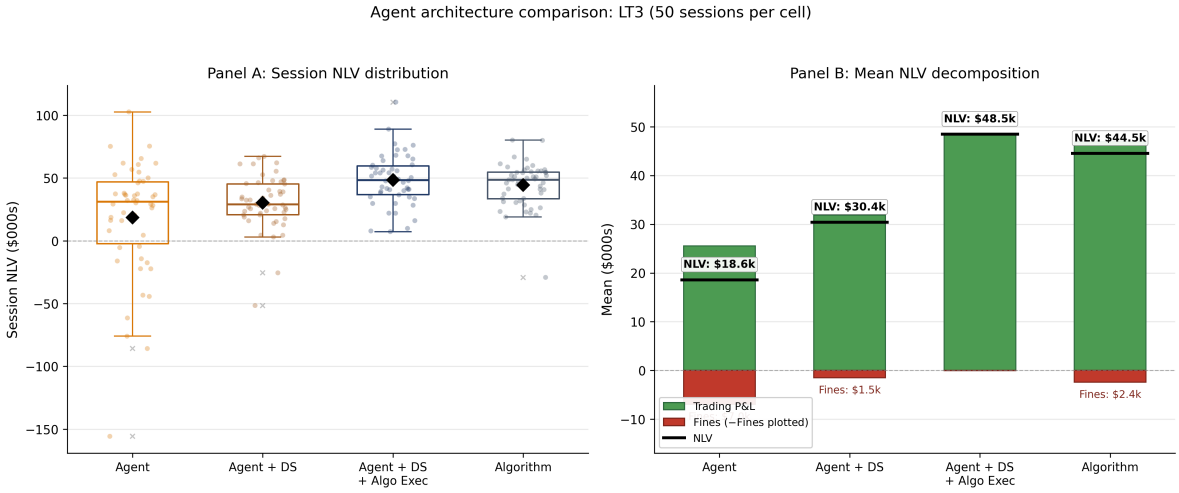


Figure 10: Connected-means view of model and reasoning (50 sessions per reasoning cell for each model; all at normal simulation speed). **Panel A**: mean session NLV ($000s); **Panel B**: mean think time per agent turn (ms, log scale). Within each model's x-slot, three points at small horizontal offsets show mean off, low, and medium in that order, connected by a thin grey line that traces each model's reasoning trajectory. A vertical separator divides the OpenAI family (gpt-5.4, mini, nano) from the Anthropic family (Opus 4.6, Sonnet 4.6).

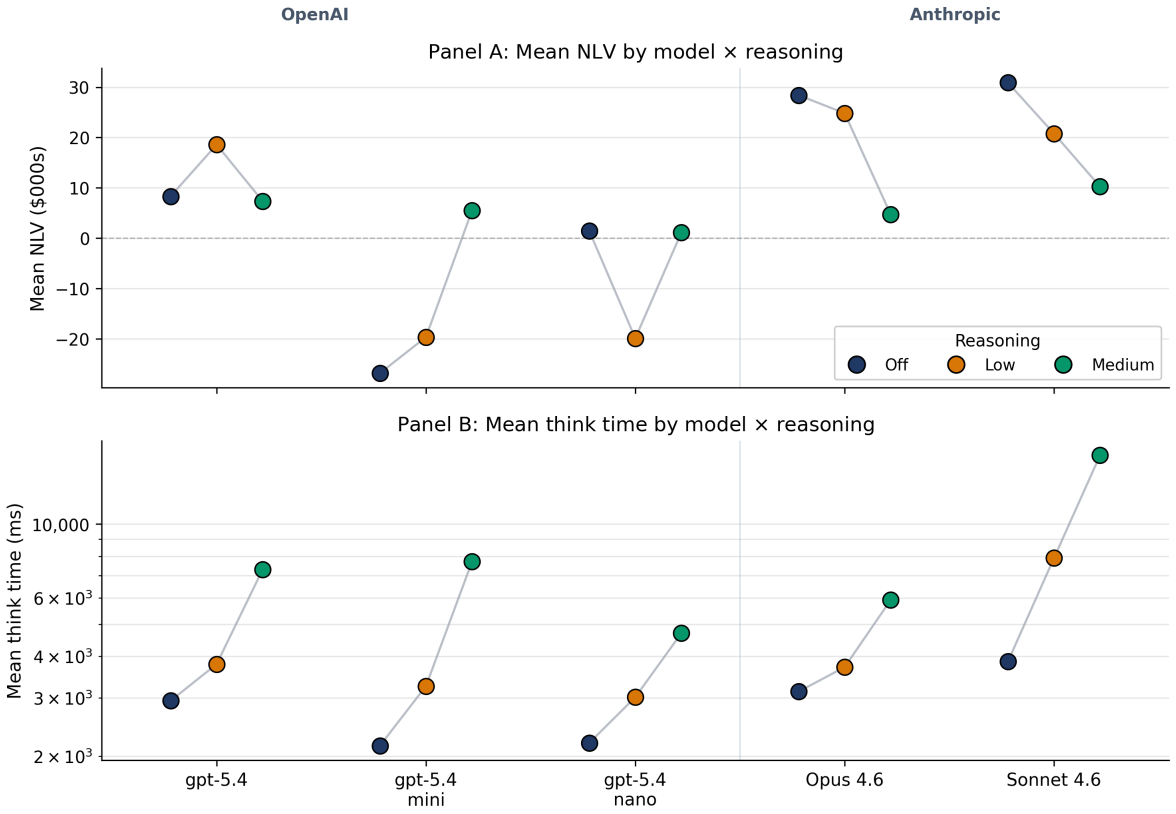


Table 1: Descriptive Treatment Means. Cell-level means across 50 gpt-5.4 sessions per cell on the reasoning × simulation-speed grid ($N = 450$). Dollar amounts in thousands ($000s). NLV is net liquidation value (Trading P&L minus Fines). Sharpe is mean NLV divided by std NLV within cell. % Profitable is the fraction of sessions with positive NLV. Think Time is the mean milliseconds per LLM turn. Tenders Accepted is the mean count of accepted tenders per session.

Cell	Mean NLV	Std NLV	Sharpe	% Profitable	Mean Trading P&L	Mean Fines	Mean Think Time (ms)	Tenders Accepted
Normal / Off	8.3	37.0	0.22	62%	22.6	14.3	2,939	3.7
Normal / Low	18.6	46.3	0.40	74%	25.5	7.0	3,789	3.7
Normal / Medium	7.3	62.8	0.12	62%	28.8	21.4	7,303	3.3
Fast / Off	10.7	40.9	0.26	72%	18.7	8.0	2,652	3.9
Fast / Low	14.0	41.2	0.34	72%	26.5	12.4	3,843	3.4
Fast / Medium	−18.4	69.0	-0.27	46%	14.5	33.0	8,372	2.5
Slow / Off	2.7	46.1	0.06	70%	10.7	8.1	3,231	3.4
Slow / Low	22.9	27.6	0.83	88%	23.1	0.2	3,728	3.6
Slow / Medium	27.8	34.9	0.80	78%	34.5	6.6	5,683	3.8
All gpt-5.4	10.4	48.1	0.22	69%	22.8	12.3	4,615	3.5

Table 2: Reasoning, Response Time, and Performance (Normal speed). OLS regressions on 150 gpt-5.4 sessions in the normal-speed market (50 per reasoning level). Dependent variables are NLV, Trading P&L, and Fines (all in $000s). Columns (1), (3), and (5) regress each DV on reasoning indicators alone (reference: reasoning off). Columns (2), (4), and (6) add log average think time per agent turn (mean-centered within the gpt-5.4 subsample). NLV equals Trading P&L minus Fines. Dollar amounts in thousands ($000s). HC3 robust standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels. $N = 150$ (50 sessions per reasoning level). Reference: reasoning off.

	(1) NLV	(2) NLV	(3) TP&L	(4) TP&L	(5) Fines	(6) Fines
Low	10.27	31.27***	2.94	18.38***	−7.33*	−12.88***
	(8.47)	(8.46)	(6.60)	(6.56)	(4.11)	(4.61)
Medium	−0.97	74.45***	6.18	61.66***	7.15	−12.79*
	(10.41)	(12.90)	(7.63)	(10.48)	(5.65)	(7.64)
ln(Think Time)		−88.91***		−65.40***		23.51***
		(14.21)		(11.82)		(7.32)
Constant	8.31	−23.83***	22.59***	−1.05	14.28***	22.78***
	(5.29)	(7.34)	(4.23)	(6.12)	(3.24)	(4.27)
N	150	150	150	150	150	150
R^2	0.0105	0.2415	0.0047	0.2328	0.0540	0.1153

Table 3: Reasoning, Market Speed, and Performance. Panel A reports pairwise speed contrasts on session NLV within each reasoning level: the joint F(speed) test, Normal−Fast, and Slow−Fast. Panel B is the analog along the reasoning axis: the joint F(reasoning) test, Medium−Low, and Off−Low. Fast and slow scale wall-clock time per tick (RIT `Speed` = 150 for fast, 100 for normal, 50 for slow); the agent poll interval scales in tandem so the poll-to-tick ratio matches the normal-speed baseline. All speed cells use gpt-5.4; 50 sessions per reasoning × speed cell. The first data column reports the joint Wald F on the two contrast indicators within each conditioning subset (df = 2, 147); the p-value is in parentheses below. The remaining columns show pairwise contrasts: each interior cell stacks the mean NLV difference (in $000s, signed, two decimals; stars from the HC3 robust t-statistic), the HC3 robust standard error in parentheses, and the Sharpe-ratio difference [ΔS] for the indicated comparison. Mean differences and standard errors come from one OLS regression of NLV on the contrast indicators within each conditioning subset, with the reference cell as fast (Panel A) or low (Panel B). The *Pooled with FE* row in each panel reports a pooled OLS over all $N = 450$ sessions with the orthogonal axis as fixed effects; the F-statistic tests the joint zero of the two contrast indicators (df = 2, 446). No Sharpe-ratio difference is reported for the pooled rows. ***, **, * denote significance at the 1%, 5%, and 10% levels. $N = 150$ per conditional row; $N = 450$ for each pooled row.

Panel A: Speed contrasts on NLV by reasoning level

	F(speed)	Normal − Fast	Slow − Fast
Off	0.43 (0.652)	−2.35 (7.88) [$\Delta S = -0.04$]	−8.00 (8.80) [$\Delta S = -0.20$]
Low	0.81 (0.446)	4.54 (8.86) [$\Delta S = 0.06$]	8.90 (7.09) [$\Delta S = 0.49$]
Medium	9.38*** (<0.001)	25.78* (13.33) [$\Delta S = 0.38$]	46.28*** (11.05) [$\Delta S = 1.07$]
Pooled w/ Reasoning FE	4.28** (0.014)	9.32 (5.97)	15.72*** (5.39)

Panel B: Reasoning contrasts on NLV by speed

	F(reasoning)	Medium − Low	Off − Low
Fast	4.22** (0.017)	−32.46*** (11.49) [$\Delta S = -0.61$]	−3.38 (8.29) [$\Delta S = -0.08$]
Normal	0.86 (0.425)	−11.23 (11.14) [$\Delta S = -0.28$]	−10.27 (8.47) [$\Delta S = -0.18$]
Slow	4.94*** (0.008)	4.91 (6.35) [$\Delta S = -0.03$]	−20.28*** (7.67) [$\Delta S = -0.77$]
Pooled w/ Speed FE	3.89** (0.021)	−12.93** (5.79)	−11.31** (4.74)

Table 4: Within-Reasoning Quantile Regressions on NLV. For each reasoning level (Off, Low, Medium), three quantile regressions of session NLV on speed indicators with fast as the reference cell, plus the q75−q25 interquartile-range coefficient. A fourth pooled sub-panel fits all $N = 450$ sessions with additive reasoning fixed effects. Companion Table 5 reports the analog along the speed axis. Fast and slow scale wall-clock time per tick (RIT `Speed` = 150 for fast, 100 for normal, 50 for slow); the agent poll interval scales in tandem so the poll-to-tick ratio matches the normal-speed baseline. All speed cells use gpt-5.4; 50 sessions per reasoning × speed cell. Quantile regressions of session NLV (in \$000s) at $\tau \in \{0.25, 0.50, 0.75\}$ with robust asymptotic SEs in parentheses. The *IQR* column reports the interquartile-range coefficient (q75 − q25) with bootstrapped SEs. ***, **, * denote significance at the 1%, 5%, and 10% levels. The first three sub-panels each fit $N = 150$ gpt-5.4 sessions (50 per cell). The final *Pooled* sub-panel fits all $N = 450$ sessions with additive reasoning fixed effects (reference: low), so each speed coefficient is the average speed contrast holding reasoning composition fixed; the constant is the predicted NLV at fast speed and low reasoning.

	q25	q50	q75	IQR
Reasoning: Off				
Normal vs Fast	−6.06	0.91	−5.67	0.38
	(14.98)	(8.67)	(6.91)	(14.00)
Slow vs Fast	5.88	−5.64	−8.23	−14.11
	(14.98)	(8.67)	(6.91)	(17.22)
Constant (Fast)	−12.74	22.09***	40.48***	53.22***
	(10.59)	(6.13)	(4.89)	(11.92)
N	150	150	150	150
Reasoning: Low				
Normal vs Fast	5.32	6.53	2.79	−2.53
	(11.40)	(6.91)	(6.46)	(18.67)
Slow vs Fast	26.53**	4.60	−2.71	−29.24**
	(11.40)	(6.91)	(6.46)	(14.60)
Constant (Fast)	−9.82	23.69***	44.49***	54.31***
	(8.06)	(4.89)	(4.57)	(13.19)
N	150	150	150	150
Reasoning: Medium				
Normal vs Fast	22.65	27.65**	25.38**	2.73
	(16.05)	(13.20)	(11.32)	(19.23)
Slow vs Fast	56.72***	44.83***	18.79*	−37.93**
	(16.05)	(13.20)	(11.32)	(16.68)
Constant (Fast)	−50.01***	−8.93	30.11***	80.12***
	(11.35)	(9.33)	(8.01)	(11.55)
N	150	150	150	150
All Reasoning Levels (Pooled, additive reasoning FE)				
Normal vs Fast	6.02	9.78*	4.57	−0.92
	(7.72)	(5.00)	(3.94)	(10.63)
Slow vs Fast	27.72***	11.44**	−1.28	−28.85***
	(7.72)	(5.00)	(3.94)	(9.12)
Constant (Fast, Low)	−10.34	20.43***	43.02***	52.81***
	(6.98)	(4.57)	(3.61)	(9.30)
N	450	450	450	450

Table 5: Within-Speed Quantile Regressions on NLV. For each speed level (Slow, Normal, Fast), three quantile regressions of session NLV on reasoning indicators with low as the reference cell, plus the q75−q25 interquartile-range coefficient. A fourth pooled sub-panel fits all $N = 450$ sessions with additive speed fixed effects. Companion Table 4 reports the analog along the reasoning axis. Fast and slow scale wall-clock time per tick (RIT `Speed` = 150 for fast, 100 for normal, 50 for slow); the agent poll interval scales in tandem so the poll-to-tick ratio matches the normal-speed baseline. All speed cells use gpt-5.4; 50 sessions per reasoning × speed cell. Quantile regressions of session NLV (in $000s) at $\tau \in \{0.25, 0.50, 0.75\}$ with robust asymptotic SEs in parentheses. The *IQR* column reports the interquartile-range coefficient (q75 − q25) with bootstrapped SEs. ***, **, * denote significance at the 1%, 5%, and 10% levels. The first three sub-panels each fit $N = 150$ gpt-5.4 sessions (50 per cell). The final *Pooled* sub-panel fits all $N = 450$ sessions with additive speed fixed effects (reference: fast), so each reasoning coefficient is the average reasoning contrast holding speed composition fixed; the constant is the predicted NLV at low reasoning and fast speed.

	q25	q50	q75	IQR
Speed: Slow				
Off vs Low	−23.57**	−11.84*	−9.53*	14.04
	(10.42)	(6.23)	(5.52)	(13.83)
Medium vs Low	−10.00	7.62	7.12	17.12
	(10.42)	(6.23)	(5.52)	(12.77)
Constant (Low)	16.71**	28.29***	41.77***	25.07***
	(7.37)	(4.41)	(3.90)	(5.77)
N	150	150	150	150
Speed: Normal				
Off vs Low	−14.29	−7.22	−12.47	1.82
	(18.09)	(11.22)	(9.18)	(14.88)
Medium vs Low	−22.85	−11.50	8.21	31.06
	(18.09)	(11.22)	(9.18)	(19.50)
Constant (Low)	−4.51	30.22***	47.28***	51.78***
	(12.79)	(7.93)	(6.49)	(12.88)
N	150	150	150	150
Speed: Fast				
Off vs Low	−2.91	−1.60	−4.01	−1.09
	(17.38)	(11.02)	(9.89)	(17.44)
Medium vs Low	−40.19**	−32.62***	−14.38	25.81
	(17.38)	(11.02)	(9.89)	(18.33)
Constant (Low)	−9.82	23.69***	44.49***	54.31***
	(12.29)	(7.79)	(6.99)	(12.80)
N	150	150	150	150
All Speed Levels (Pooled, additive speed FE)				
Off vs Low	−14.48*	−7.23	−7.89**	7.13
	(7.72)	(5.00)	(3.94)	(7.88)
Medium vs Low	−23.21***	−9.42*	2.30	25.89***
	(7.69)	(5.00)	(3.94)	(9.58)
Constant (Low, Fast)	−10.34	20.43***	43.02***	52.81***
	(6.98)	(4.57)	(3.61)	(9.37)
N	450	450	450	450

Table 6: Tender Selection and Responsiveness. Panel A reports cell means on the per-turn live sample (one row per agent turn while the tender is alive, pre-decision). *Book Edge | Intent* and *Book Edge | No Intent* ($/share) are cell means of book edge restricted to accept and decline turns respectively. *Neg-Book-Edge Intent* is the share of accept turns with book edge < 0. The right-most column, *Intent Failed*, is the only per-tender statistic: share of accept intents whose server submission did not produce a valid fill (tender expired during the agent's think time, or implied post-fill position would have breached the net limit). Matches the corresponding column in Table 7 Panel A. Panel B: cols (1)–(2) regress accept intent on book edge with reasoning indicators on the per-turn sample, col (2) adding reasoning × book edge; col (3) regresses the Intent Failed indicator on reasoning across the full per-tender accept-intent sample. Of 71 intent failures, 52 expired in think time and 19 would have hit the position limit. All regressions include speed FE; cluster-robust SEs (by session) in parentheses. ***/**/* = 1/5/10%.

Panel A: Summary statistics (live tender turns)

Cell	Book Edge | Intent ($/share)	Book Edge | No Intent ($/share)	Neg-Book-Edge Intent	Intent Failed
Normal / Off	0.1120	0.0520	10%	3%
Normal / Low	0.1262	-0.0588	2%	1%
Normal / Medium	0.1290	-0.0201	1%	11%
Fast / Off	0.1129	0.0559	9%	3%
Fast / Low	0.1346	-0.0306	2%	3%
Fast / Medium	0.1367	-0.0530	2%	16%
Slow / Off	0.1196	0.0555	7%	1%
Slow / Low	0.1244	-0.0617	2%	2%
Slow / Medium	0.1244	-0.0849	2%	2%

Panel B: Selection and intent-failure regressions

	(1) Intent	(2) Intent	(3) Intent Failed
Low	0.2381***	0.1252***	−0.0027
	(0.0336)	(0.0315)	(0.0087)
Medium	0.3290***	0.1995***	0.0714***
	(0.0311)	(0.0291)	(0.0139)
Book Edge	1.6073***	0.6834***	
	(0.1607)	(0.1815)	
Low × Book Edge		1.8507***	
		(0.2175)	
Medium × Book Edge		1.8265***	
		(0.2101)	
Constant	0.2632***	0.3253***	0.0234**
	(0.0284)	(0.0291)	(0.0094)
Speed FE	Yes	Yes	Yes
Sample	Turn	Turn	Tender
N	3060	3060	1644
R^2	0.2723	0.3338	0.0389

Table 7: Tender Execution Quality. The sample for this table is the set of tenders for which the agent issued an accept action and that accept was filled. Panel A reports per-cell summary statistics on the post-fill unwind. *Unwind Completed* is the share of filled intents for which the agent reached a flat position before session end; the complementary share is *stuck* (residual inventory remained at session end and was auto-liquidated). *Edge Captured* ($/share) is the mean realized edge per share, restricted to completed unwinds. *Book Edge at Intent* ($/share) is the mean book edge at the decision turn, restricted to the same completed-unwind tenders for direct comparison with Edge Captured. *IS* ($/share) is the per-share implementation shortfall on the filled-intent sample: $\frac{1}{Q}[-s(P_t - \pi_0)Q + s(\bar{p}_u - \pi_0)u + s(\pi_1 - \pi_0)r + 0.02u + 1.00r]$, where $s = +1$ for SELL tenders / -1 for BUY, P_t is the tender price, π_0 the bid-ask midpoint at accept, π_1 the bid-ask midpoint at settlement, $\bar{p}_u$ the unwind VWAP, u the shares unwound, and $r = Q - u$ the residual auto-liquidated at session end. Components are tender capture, execution cost, opportunity cost, $0.02/share trading fees, and $1/share auto-liquidation fines. Positive IS = drag on realized P&L vs the paper-portfolio benchmark. Panel B reports the corresponding cell-level regressions on reasoning indicators (reference: off) with speed fixed effects. Col (1) is a linear probability model on the full filled-intent sample. Col (2) regresses Edge Captured on reasoning controlling for book edge at the decision turn (completed unwinds). Col (3) regresses Book Edge at Intent on reasoning (completed unwinds). Col (4) regresses IS on reasoning controlling for book edge at the decision turn (filled intents); its sample is smaller than col (1) because the book-edge control is undefined for filled intents whose visible book at the decision turn could not span the full tender quantity. A negative coefficient implies lower per-share shortfall, or realized P&L closer to the paper-portfolio benchmark. Cluster-robust standard errors (by session) in parentheses. ***/**/* = 1/5/10%.

Panel A: Summary statistics (filled intents)

Cell	N	Unwind Completed	Edge Captured ($/share)	Book Edge at Intent ($/share)	IS ($/share)
Normal / Off	187	88%	0.1098	0.1124	-0.0236
Normal / Low	187	89%	0.1396	0.1307	-0.0366
Normal / Medium	166	78%	0.1582	0.1327	0.0025
Fast / Off	194	88%	0.0981	0.1151	-0.0221
Fast / Low	169	83%	0.1513	0.1336	-0.0279
Fast / Medium	125	68%	0.1430	0.1371	0.1576
Slow / Off	172	92%	0.0801	0.1247	-0.0105
Slow / Low	181	97%	0.1065	0.1256	-0.0871
Slow / Medium	192	91%	0.1582	0.1301	-0.0902

Panel B: Cell-level regressions

	(1) Unwind Completed	(2) Edge Captured	(3) Book Edge at Intent	(4) IS
Low	0.0003	0.0242**	0.0126**	−0.0137
	(0.0223)	(0.0119)	(0.0049)	(0.0211)
Medium	−0.0977***	0.0470**	0.0155***	0.0647**
	(0.0252)	(0.0183)	(0.0051)	(0.0296)
Book edge		0.8935***		−1.2054***
		(0.0916)		(0.1205)
Constant	0.8818***	0.0006	0.1159***	0.0947***
	(0.0219)	(0.0176)	(0.0051)	(0.0264)
Speed FE	Yes	Yes	Yes	Yes
Sample	Filled	Completed	Completed	Filled
N	1573	1231	1231	1414
R^2	0.0411	0.0981	0.0088	0.0716

Table 8: Reasoning Content Classification (Normal speed). Per-session reasoning content was labeled along eight binary dimensions by an LLM classifier from a different model family than the subject models; five dimensions are reported here. Panel A shows cell-level means across the three gpt-5.4 normal-speed reasoning cells (50 sessions per cell, $N = 150$). *Time-Aware* and *Pos-Limit Aware*: fraction of all turns in which reasoning explicitly invokes remaining time or the position limit. *Unwind Feas.*: fraction of tender-visible turns discussing post-accept unwind feasibility. *Aggressive Unwind* and *Patient Waiting*: fraction of nonzero-position turns reasoning with urgency to flatten or with deliberate patience, respectively. An eighth dimension (*Inaction Holding*, the fraction of nonzero-position turns with no orders, cancels, or tender decisions) is omitted because it correlates with *Patient Waiting* at the session level ($r = 0.85$). Panel B reports OLS regressions of each classifier dimension on reasoning indicators (reference: reasoning off), restricted to the 150 gpt-5.4 normal-speed sessions. HC3 robust standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Cell means of classifier dimensions

Cell	Time Aware	Pos-Limit Aware	Unwind Feas.	Aggressive Unwind	Patient Waiting
Normal / Off	0.08	0.01	0.37	0.17	0.19
Normal / Low	0.11	0.01	0.61	0.30	0.19
Normal / Medium	0.18	0.03	0.66	0.33	0.15
All gpt-5.4 normal	0.12	0.01	0.55	0.27	0.18

Panel B: Reasoning level → classifier dimensions

	(1) Time Aware	(2) Pos-Limit Aware	(3) Unwind Feas.	(4) Aggressive Unwind	(5) Patient Waiting
Low	0.034***	0.003	0.244***	0.129**	0.002
	(0.012)	(0.004)	(0.058)	(0.056)	(0.043)
Medium	0.099***	0.020***	0.294***	0.156***	−0.037
	(0.015)	(0.007)	(0.058)	(0.054)	(0.039)
Constant	0.077***	0.007**	0.370***	0.172***	0.189***
	(0.008)	(0.003)	(0.051)	(0.046)	(0.032)
N	150	150	150	150	150
R^2	0.254	0.076	0.198	0.070	0.009

Table 9: Agent Architecture Comparison. Four architectures applied to the LT3 case, 50 sessions per cell. DS denotes *decision support*: pre-computed book-edge, inventory, limit, and flattening-cost summaries. Algo Exec denotes *algorithmic execution* using the fixed reduce-only unwind routine. Algorithm denotes the fully compiled benchmark with no LLM calls. Panel A reports per-cell descriptive means using the columns of Table 1: *Mean / Std NLV* are the cross-session mean and standard deviation of session NLV; *Sharpe* is mean / std; *% Profitable* is the share of sessions with positive NLV; *Mean Trading P&L* and *Mean Fines* are the components of NLV; *Mean Control Time* is the average wall-clock spent per call by the agent's controlling script, in milliseconds. For the Agent and Agent + DS rows, this equals the mean LLM-call latency since every call goes to the LLM. For the Agent + DS + Algo Exec and Algorithm rows, this includes the near-zero latency during idle turns and latency in turns where the script acts (fast accept/declines/unwinds from the algorithm or relatively slower think time from pending LLM decisions). The two Algorithm rows differ only in the tender-acceptance threshold ($0.05 and $0.10 per share); all other algorithm parameters and the case configuration are identical. *Tenders Accepted* is the mean count of accepted tenders per session. Panel B reports regressions on the pooled sample ($N = 200$) of the four cells: Agent / Low, Agent + DS / Low, Agent + DS + Algo Exec / Medium, and the Algorithm ($0.05 threshold). Three architecture indicators are included with Agent / Low as the omitted base, so Constant equals the Agent / Low cell mean and each indicator coefficient is the mean difference between that architecture and the Agent / Low cell. Columns: (1) NLV OLS, (2) NLV IQR ($q_{75} - q_{25}$), (3) Trading P&L OLS, (4) Fines OLS. Dollar amounts in thousands ($000s). HC3 robust standard errors for OLS in parentheses; bootstrap SEs for IQR. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Descriptive means

Cell	Mean NLV	Std NLV	Sharpe	% Profitable	Mean Trading P&L	Mean Fines	Mean Control Time (ms)	Tenders Accepted
Agent								
Off	8.3	37.0	0.22	62%	22.6	14.3	2,939	3.7
Low	18.6	46.3	0.40	74%	25.5	7.0	3,789	3.7
Medium	7.3	62.8	0.12	62%	28.8	21.4	7,303	3.3
Agent + DS								
Off	19.6	31.1	0.63	78%	30.6	11.1	2,860	4.1
Low	30.4	21.5	1.41	96%	31.9	1.5	2,917	4.1
Medium	26.9	42.9	0.63	84%	31.8	4.8	5,196	3.5
Agent + DS + Algo Exec								
Off	44.5	15.6	2.85	100%	45.6	1.1	168	4.4
Low	38.0	25.5	1.49	92%	38.2	0.2	673	4.5
Medium	48.5	20.4	2.38	100%	48.5	0.0	1,387	4.6
Algorithm								
$0.05 threshold	44.5	17.9	2.49	98%	46.9	2.4	25	3.9
$0.10 threshold	44.0	24.9	1.77	96%	48.6	4.6	37	3.4

Panel B: Regressions on architecture indicators

	(1) NLV (OLS)	(2) NLV (IQR)	(3) TP&L (OLS)	(4) Fines (OLS)
Agent + DS	11.82	−26.86*	6.35	−5.47**
	(7.29)	(14.19)	(5.89)	(2.60)
Agent + DS + Algo Exec	29.88***	−28.18**	22.96***	−6.91***
	(7.23)	(14.29)	(5.84)	(2.52)
Algorithm	25.93***	−30.46**	21.39***	−4.53
	(7.09)	(14.26)	(5.50)	(2.92)
Constant (Agent / Low)	18.58***	51.78***	25.53***	6.95***
	(6.61)	(13.56)	(5.06)	(2.52)
N	200	200	200	200
R^2	0.1463		0.1429	0.0588

Table 10: Provider Robustness: Descriptive Means Across Models. Cell-level means across 50 sessions per (model × reasoning) cell, extending Table 1 to four additional models: two OpenAI scales (gpt-5.4-mini, gpt-5.4-nano) and two Anthropic models (claude-opus-4-6, claude-sonnet-4-6). Low and Medium reasoning labels correspond to OpenAI `reasoning_effort` or Anthropic extended-thinking budgets, depending on the provider. Same columns as Table 1; normal simulation speed only ($N = 750$). Sharpe is mean NLV divided by std NLV within cell. Dollar amounts in thousands ($000s).

Cell	Mean NLV	Std NLV	Sharpe	% Profitable	Mean Trading P&L	Mean Fines	Mean Think Time (ms)	Tenders Accepted
gpt-5.4 / Off	8.3	37.0	0.22	62%	22.6	14.3	2,939	3.7
gpt-5.4 / Low	18.6	46.3	0.40	74%	25.5	7.0	3,789	3.7
gpt-5.4 / Medium	7.3	62.8	0.12	62%	28.8	21.4	7,303	3.3
gpt-5.4-mini / Off	−26.8	54.5	-0.49	40%	9.8	36.7	2,153	2.7
gpt-5.4-mini / Low	−19.7	45.7	-0.43	42%	8.7	28.4	3,255	4.5
gpt-5.4-mini / Medium	5.5	59.9	0.09	64%	24.1	18.6	7,709	3.7
gpt-5.4-nano / Off	1.4	19.7	0.07	36%	5.6	4.2	2,195	1.3
gpt-5.4-nano / Low	−19.9	49.0	-0.41	42%	10.0	29.9	3,014	3.1
gpt-5.4-nano / Medium	1.1	48.2	0.02	56%	19.1	18.0	4,695	3.9
Opus 4.6 / Off	28.4	25.2	1.13	86%	29.3	0.9	3,132	4.0
Opus 4.6 / Low	24.8	31.4	0.79	80%	27.0	2.2	3,712	3.8
Opus 4.6 / Medium	4.7	54.5	0.09	60%	16.5	11.8	5,911	3.7
Sonnet 4.6 / Off	30.9	35.2	0.88	80%	35.4	4.5	3,859	4.3
Sonnet 4.6 / Low	20.8	32.2	0.64	76%	34.6	13.8	7,904	3.5
Sonnet 4.6 / Medium	10.2	39.3	0.26	62%	19.2	9.0	16,107	3.1

Table 11: NN and LLM Tender Selection Across Liquidity Conditions. Means across 50 independent sessions per cell; dollar amounts in $000s. Normal, half, and quarter frequency correspond to 120, 60, and 30 computer-trader orders per simulation minute, with the market clock fixed. Both selectors use the same NN execution policy; the LLM is `gpt-5.4` with medium reasoning. Sharpe is mean NLV divided by its standard deviation. Panels B and C test changes from normal frequency using Welch tests, with standard errors in parentheses.

Panel A: Descriptive statistics

Cell	Mean NLV	Std NLV	Sharpe	% Profitable	Mean Trading P&L	Mean Fines	Tenders Accepted
NN trader / Normal	42.1	19.2	2.20	96%	42.1	0.0	2.2
NN trader / Half	31.6	20.8	1.52	86%	31.6	0.0	1.7
NN trader / Quarter	15.5	15.7	0.98	58%	15.5	0.0	1.0
LLM agent / Normal	42.8	16.8	2.55	100%	42.8	0.0	4.3
LLM agent / Half	37.6	19.0	1.98	98%	37.6	0.0	3.7
LLM agent / Quarter	23.9	10.7	2.23	98%	23.9	0.0	3.1

Panel B: Normal to half frequency

Trader	Normal	Half	Change (SE)	t	p
NN trader	42.1	31.6	−10.5 (4.00)	−2.62	0.010
LLM agent	42.8	37.6	−5.2 (3.58)	−1.46	0.148

Panel C: Normal to quarter frequency

Trader	Normal	Quarter	Change (SE)	t	p
NN trader	42.1	15.5	−26.7 (3.51)	−7.60	< 0.001
LLM agent	42.8	23.9	−19.0 (2.82)	−6.73	< 0.001

Online Appendix: Tables

Table A1: Case Parameters. Case parameters for the LT3 liability trading experiment. All sessions use the same case configuration with randomized price paths.

Parameter	Value
Market structure	
Securities	CRZY, TAME (two independent order books)
Duration	300 ticks
Max order size	CRZY: 25,000 shares; TAME: 10,000 shares
Position limits	
Net limit	100,000 shares
Gross limit	250,000 shares
Tender offers	
Arrival	Institutional tenders arrive throughout the session
Duration	$\approx$ 30 ticks per tender
Costs and penalties	
Transaction fee	\$0.02 per share (market trades only; tender fills free)
Front-running fine	\$0.20/share $\leq$ 5,000 shares; \$0.40/share beyond
End-of-case fine	\$1.00/share on uncovered positions
Experimental design	
Model	gpt-5.4-2026-03-05
Reasoning levels	Off, Low, Medium
Speeds	50%, 100%, 150%
Sessions per cell	50
Total sessions	450 (9 cells $\times$ 50)
Agent polling	$\approx$ 1 tick between observations

Table A2: LT3 Prompt: Rendered Example. System prompt and user message for the first tender-visible turn of an example LT3 session.

```
=== SYSTEM PROMPT ===
You are a liability trader in a live RIT (Rotman Interactive Trader) case.

## Securities
There are 2 securities: CRZY, TAME.
Each has its own order book. When submitting orders, use the ticker as the "venue" parameter (e.g. venue="CRZY").
Positions are tracked per security.

**Order size limits** (per single order -- RIT will REJECT orders exceeding these):
  CRZY: max 25,000 shares per order
  TAME: max 10,000 shares per order
If you need to trade more than the limit, split into multiple smaller orders.

## Case Structure
Duration: 300 ticks (each tick is ~1 second of real time).
You are called periodically to observe and optionally act. The tick numbers in your observations show how much time has elapsed between your
      turns.

## Your Job: Liability Trading
Institutional tender offers arrive during the case. Each tender specifies a ticker, a side (BUY or SELL from your perspective), a quantity, a
      price, and a deadline.

Your job:
1. Evaluate the tender: check the order book depth for THAT ticker. Can you unwind profitably?
2. If yes, ACCEPT the tender using decide_tender.
3. Unwind the acquired position by submitting orders on the SAME ticker.
4. Tenders remain active for approximately 30 ticks (~30 seconds). Due to polling latency, you may not see a tender until a few ticks after
      it is issued. The time_left field shows how many ticks remain before expiration. If you do not accept before the deadline, the tender
      expires automatically -- there is no need to explicitly decline.

Do NOT speculate or make markets. Only trade to unwind accepted tenders.

## Front-Running Prohibition
Do NOT trade a security to anticipate a pending tender. If you have no existing position in a security, you must accept or let expire the
      tender BEFORE placing any orders on that security. Exception: if you are already unwinding a position from a previously accepted
      tender, you may continue closing that position even if a new tender on the same security arrives.

## Position Limits
Net limit: 100,000 shares (sum of all positions)
Gross limit: 250,000 shares (sum of absolute positions)

## Transaction Fees
Fee: $0.02 per share on every market trade.
Tender fills do not incur a transaction fee.

## Fines
Front-running or speculative trading (any trade not directly unwinding an accepted tender) incurs fines: $0.20/share up to 5,000 shares,
      $0.40/share beyond that. Uncovered positions at the end of the case are fined at $1.00/share. These fines are assessed in real time and
```

Table A2 *(continued)*

```
      reduce your P&L.

**Important:** If you submit unwind orders in the same action as accepting a tender, you may incur front-running fines because the tender has
      not yet settled. Accepting the tender in one step then submitting unwind orders in a subsequent step will avoid these fines.

## End of Case
Aim to be flat (zero inventory on all securities) before the case ends. Monitor the tick count and close positions early.

## Available Tools
- submit_order: Submit an order. Set venue to the ticker name. Types: LIMIT, MARKETABLE_LIMIT, MARKET.
- cancel_order: Cancel a resting order by order_id.
- decide_tender: Accept or decline a tender by tender_id.
You do not need to act every turn.




## Objective
Maximise P&L by accepting profitable tenders and unwinding positions efficiently. Manage risk and close all positions before the case ends.




=== USER MESSAGE (turn 10) ===
=== Tick 27 ===

MARKET STATE (CRZY):
  Best bid: 10.10
  Best ask: 10.12
  Last: 10.12
  Volume: 88,100 My volume: 88,100
  ORDER BOOK:
    BIDS ASKS
       10.10 x 5200 10.12 x 500
       10.07 x 4500 10.13 x 5900
       10.07 x 4400 10.13 x 4600
       10.06 x 3700 10.13 x 4700
       10.06 x 5000 10.14 x 5600
       10.06 x 4500 10.16 x 5200
       10.03 x 4500 10.21 x 5600
       10.02 x 3200 10.27 x 4800
       10.02 x 4900 10.30 x 4100
       10.01 x 5500 10.32 x 5100
       10.01 x 5100 10.34 x 4400
       10.00 x 5600 10.44 x 4200
        9.99 x 5000 10.45 x 4400
        9.98 x 700
        9.98 x 4600
        9.98 x 5500
        9.97 x 4800
        9.97 x 5200
        9.97 x 5600
        9.96 x 4000

MARKET STATE (TAME):
```

Table A2 ***(continued)***

```
  Best bid: 25.15
  Best ask: 25.21
  Last: 25.15
  Volume: 101,300 My volume: 101,300
  ORDER BOOK:
    BIDS ASKS
       25.15 x 100 25.21 x 2500
       25.15 x 5700 25.27 x 4400
       25.14 x 200 25.27 x 5300
       25.11 x 4400 25.28 x 4500
       25.08 x 300 25.31 x 5700
       25.03 x 5900 25.41 x 5800
       25.02 x 5100 25.42 x 5200
       25.02 x 4400 25.43 x 4500
       25.00 x 5200 25.46 x 5500
       25.00 x 4400 25.46 x 5900
       24.97 x 300 25.51 x 4600
       24.95 x 5100 25.58 x 4600
       24.95 x 5700 25.63 x 4600
       24.94 x 6000
       24.92 x 5700
       24.92 x 5700
       24.92 x 5900
       24.92 x 5400
       24.92 x 5600
       24.92 x 5500

YOUR POSITIONS:
  CRZY: 0 shares (max order: 25000) P&L: +0.00
  TAME: 0 shares (max order: 10000) P&L: +0.00
  Net: 0 shares

YOUR P&L: $0.00
YOUR FINES: $0.00
NLV (net of fines): $0.00

YOUR LIMITS:
  Net: 0 / 100,000 (100,000 remaining)
  Gross: 0 / 250,000 (250,000 remaining)

TENDERS:
  [0] "An institution would like to BUY 70000 shares of CRZY from you at a price of 10.46. Would you like to SELL the shares to them?"
        (expires in 29 ticks)
```

Table A3: Measurement robustness: fixed-window edge capture (W30). Mirrors Table 7 (Tender Execution Quality) on the same intent-funnel frame, with the only substantive change being that the fixed-window measure *W30* replaces *Edge Captured* in Panel A column (3) and as the dependent variable in Panel B column (2). *W30* is the ticker PnL delta between the accept turn and a deterministic endpoint 30 ticks later (capped at the next same-ticker accept or session end), divided by tender quantity. Because the window length is fixed rather than scaling with unwind duration, W30 holds exposure constant across cells – an alternative to *Edge Captured* that does not absorb variation in unwind duration. Panel A reports per-cell summary statistics on the post-fill unwind. *Unwind Completed* is the share of filled intents for which the agent reached a flat position before session end. *W30 Captured* ($/share) is the cell mean of W30 restricted to completed unwinds; *Book Edge at Intent* ($/share) is the cell mean of the book edge at the decision turn restricted to the same matched completed-unwind sub-sample (so the W30 column and the Book Edge at Intent column compare on identical tenders). Panel B reports the corresponding cell-level regressions on reasoning indicators (reference: off) with speed fixed effects. Col (1) is a linear probability model on the full filled-intent sample. Col (2) regresses W30 on reasoning controlling for book edge at the decision turn (completed unwinds). Col (3) regresses Book Edge at Intent on reasoning (completed unwinds). Cluster-robust standard errors (by session) in parentheses. ***/**/* = 1/5/10%.

Panel A: Summary statistics (filled intents)

Cell	N	Unwind Completed	W30 Captured ($/share)	Book Edge at Intent ($/share)
Normal / Off	187	88%	0.1339	0.1124
Normal / Low	187	89%	0.1589	0.1307
Normal / Medium	166	78%	0.2119	0.1327
Fast / Off	194	88%	0.1338	0.1151
Fast / Low	169	83%	0.1819	0.1336
Fast / Medium	125	68%	0.2467	0.1371
Slow / Off	172	92%	0.1066	0.1247
Slow / Low	181	97%	0.1136	0.1256
Slow / Medium	192	91%	0.1700	0.1301

Panel B: Cell-level regressions

	(1) Unwind Completed	(2) W30 Captured	(3) Book Edge at Intent
Low	0.0003	0.0164**	0.0126**
	(0.0223)	(0.0079)	(0.0049)
Medium	−0.0977***	0.0701***	0.0155***
	(0.0252)	(0.0100)	(0.0051)
Book edge		0.8181***	
		(0.0640)	
Constant	0.8818***	0.0371***	0.1159***
	(0.0219)	(0.0108)	(0.0051)
Speed FE	Yes	Yes	Yes
Sample	Filled	Completed	Completed
N	1573	1231	1231
R^2	0.0411	0.2480	0.0088

Table A4: LT3 Decision-Support Prompt: Rendered Example. System prompt and user message for the first tender-visible turn of an example decision-support session; the decision-support data feeds the Agent + DS rows of Table 9.

```
=== SYSTEM PROMPT ===
You are a liability trader in a live RIT (Rotman Interactive Trader) case.

## Securities
There are 2 securities: CRZY, TAME.
Each has its own order book. When submitting orders, use the ticker as the "venue" parameter (e.g. venue="CRZY").
Positions are tracked per security.

**Order size limits** (per single order -- RIT will REJECT orders exceeding these):
  CRZY: max 25,000 shares per order
  TAME: max 10,000 shares per order
If you need to trade more than the limit, split into multiple smaller orders.

## Case Structure
Duration: 300 ticks (each tick is ~1 second of real time).
You are called periodically to observe and optionally act. The tick numbers in your observations show how much time has elapsed between your
      turns.

## Your Job: Liability Trading
Institutional tender offers arrive during the case. Each tender specifies a ticker, a side (BUY or SELL from your perspective), a quantity, a
      price, and a deadline.

Your job:
1. Evaluate the tender: check the order book depth for THAT ticker. Can you unwind profitably?
2. If yes, ACCEPT the tender using decide_tender.
3. Unwind the acquired position by submitting orders on the SAME ticker.
4. Tenders remain active for approximately 30 ticks (~30 seconds). Due to polling latency, you may not see a tender until a few ticks after
      it is issued. The time_left field shows how many ticks remain before expiration. If you do not accept before the deadline, the tender
      expires automatically -- there is no need to explicitly decline.

Do NOT speculate or make markets. Only trade to unwind accepted tenders.

## Front-Running Prohibition
Do NOT trade a security to anticipate a pending tender. If you have no existing position in a security, you must accept or let expire the
      tender BEFORE placing any orders on that security. Exception: if you are already unwinding a position from a previously accepted
      tender, you may continue closing that position even if a new tender on the same security arrives.

## Position Limits
Net limit: 100,000 shares (sum of all positions)
Gross limit: 250,000 shares (sum of absolute positions)

## Transaction Fees
Fee: $0.02 per share on every market trade.
Tender fills do not incur a transaction fee.

## Fines
Front-running or speculative trading (any trade not directly unwinding an accepted tender) incurs fines: $0.20/share up to 5,000 shares,
```

Table A4 ***(continued)***

```
      $0.40/share beyond that. Uncovered positions at the end of the case are fined at $1.00/share. These fines are assessed in real time and
      reduce your P&L.

**Important:** If you submit unwind orders in the same action as accepting a tender, you may incur front-running fines because the tender has
      not yet settled. Accepting the tender in one step then submitting unwind orders in a subsequent step will avoid these fines.

## End of Case
Aim to be flat (zero inventory on all securities) before the case ends. Monitor the tick count and close positions early.

## Available Tools
- submit_order: Submit an order. Set venue to the ticker name. Types: LIMIT, MARKETABLE_LIMIT, MARKET.
- cancel_order: Cancel a resting order by order_id.
- decide_tender: Accept or decline a tender by tender_id.
You do not need to act every turn.

## Decision Support
Observations may include a DECISION SUPPORT block with precomputed summaries derived from the current book, fees, and limits. These summaries
      are informational only; you remain responsible for the trading decision.



## Objective
Maximise P&L by accepting profitable tenders and unwinding positions efficiently. Manage risk and close all positions before the case ends.




=== USER MESSAGE (turn 7) ===
=== Tick 20 ===

MARKET STATE (CRZY):
  Best bid: 9.89
  Best ask: 9.94
  Last: 9.89
  Volume: 80,700 My volume: 80,700
  ORDER BOOK:
    BIDS ASKS
       9.89 x 2200 9.94 x 4800
       9.89 x 5900 9.98 x 3600
       9.88 x 4500 9.98 x 5700
       9.88 x 4200 9.98 x 4500
       9.88 x 4800 9.99 x 4500
       9.88 x 4700 10.01 x 4900
       9.84 x 5600 10.01 x 5400
       9.84 x 4100 10.03 x 4300
       9.83 x 5200 10.03 x 5400
       9.83 x 4500 10.04 x 300
       9.79 x 5900 10.08 x 4900
       9.78 x 4300 10.09 x 4500
       9.73 x 5400 10.11 x 4900
       9.73 x 4100 10.13 x 4800
       9.73 x 6000 10.17 x 4800
       9.59 x 4200 10.19 x 4400
```

Table A4 ***(continued)***

```
       9.58 x 4700 10.19 x 5700
       9.54 x 5700 10.19 x 4400
       9.51 x 5100 10.27 x 4500
                           10.27 x 5800

MARKET STATE (TAME):
  Best bid: 24.89
  Best ask: 24.91
  Last: 24.89
  Volume: 79,900 My volume: 79,900
  ORDER BOOK:
    BIDS ASKS
       24.89 x 1800 24.91 x 3900
       24.88 x 5500 24.93 x 5900
       24.87 x 5500 24.94 x 4800
       24.86 x 5600 24.97 x 5000
       24.85 x 5700 24.99 x 3600
       24.84 x 5900 25.00 x 500
       24.81 x 4900 25.02 x 5800
       24.80 x 4700 25.02 x 4600
       24.75 x 4400 25.03 x 6000
       24.75 x 5300 25.03 x 5100
       24.74 x 5200 25.03 x 4200
       24.67 x 6000 25.04 x 4800
       24.55 x 5000 25.05 x 5400
                            25.07 x 5300
                            25.11 x 5300
                            25.11 x 4600
                            25.11 x 5200
                            25.12 x 5300
                            25.16 x 5600
                            25.19 x 5500

YOUR POSITIONS:
  CRZY: 0 shares (max order: 25000) P&L: +0.00
  TAME: 0 shares (max order: 10000) P&L: +0.00
  Net: 0 shares

YOUR P&L: $0.00
YOUR FINES: $0.00
NLV (net of fines): $0.00

YOUR LIMITS:
  Net: 0 / 100,000 (100,000 remaining)
  Gross: 0 / 250,000 (250,000 remaining)

TENDERS:
  [0] "An institution would like to BUY 65000 shares of TAME from you at a price of 25.14. Would you like to SELL the shares to them?"
        (expires in 30 ticks)

DECISION SUPPORT:
  Tender #0 (TAME):
    Unwind market VWAP (full size): 25.0042
```

Table A4 ***(continued)***

```
Implied edge / share (net of fee): +0.1158
Implied total edge: +7,527.00
Projected position (TAME): -65,000
Projected net (signed): -65,000
Projected gross: 65,000
Would breach net limit: no
Would breach gross limit: no
```

Table A5: Execution-Support Prompt: Rendered Example. System prompt and user message for the first tender-visible LLM dispatch of an example execution-support session; the execution-support data feeds the Agent + DS + Algo Exec rows of Table 9.

```
=== SYSTEM PROMPT ===
You are the tender-acceptance gate in a hybrid trading system on a live RIT (Rotman Interactive Trader) liability case.

A scripted unwind algorithm runs continuously and is responsible for closing any inventory you cause to appear. Your job is narrow: when a
      tender arrives, decide whether to ACCEPT it or render NO_DECISION.

## Securities
There are 2 securities: CRZY, TAME.
Each has its own order book. Positions are tracked per security.

**Order size limits** (per single order -- RIT will REJECT orders exceeding these):
  CRZY: max 25,000 shares per order
  TAME: max 10,000 shares per order
If you need to trade more than the limit, split into multiple smaller orders.

## Case Structure
Duration: 300 ticks (each tick is ~1 second of real time).
You will be called repeatedly while a tender is live with a fresh snapshot of the market. Each call is independent. You may be called again.

## The Unwind Algorithm
If you ACCEPT a tender, an external algorithm will unwind the resulting position by submitting MARKET orders, capped at 5,000 shares per
      ticker per tick, polling once per second until the position is flat. The algorithm:
- Uses MARKET orders only -- it does NOT use LIMIT or MARKETABLE_LIMIT.
- Does NOT wait for favorable fills, rest passively, or cancel and re-price.
- Walks the visible book on the unwind side at whatever prices are available.
- Never modifies its rate based on the spread; it always burns through 5,000 shares/tick until done.

## Your Job: Tender decision
Institutional tender offers arrive during the case. Each tender specifies a ticker, a side (BUY or SELL from your perspective), a quantity, a
      price, and a deadline. You will be called to either ACCEPT or return NO_DECISION.

Your job:
1. Evaluate the tender: check the order book depth for THAT ticker. Will you profit from the trade, knowing how the unwind algorithm behaves?
2. If yes, ACCEPT the tender using tender_decision.
3. Tenders remain active for approximately 30 ticks (~30 seconds). Due to polling latency, you may not see a tender until a few ticks after
      it is issued. The time_left field shows how many ticks remain before expiration. If you do not accept before the deadline, the tender
      expires automatically -- there is no need to explicitly decline.

## Position Limits
Net limit: 100,000 shares (sum of all positions)
Gross limit: 250,000 shares (sum of absolute positions)

## Transaction Fees
Fee: $0.02 per share on every market trade.
Tender fills do not incur a transaction fee.

## Fines
```

Table A5 ***(continued)***

```
Uncovered positions at the end of the case are fined at $1.00/share.

## End of Case
Aim to be flat (zero inventory on all securities) before the case ends. Monitor the tick count.

## Available Tools
You must output your decision via the ‘tender_decision‘ tool with one of two values:
- **ACCEPT** -- bind the tender immediately. The system will execute the accept and the unwind algorithm will take over.
- **NO_DECISION** -- defer. You will likely be asked again about this same tender on a later firing as the book evolves. Tenders that are
      never accepted expire automatically; there is no need to issue an explicit decline.

Call the ‘tender_decision‘ tool exactly once with ‘{"decision": "ACCEPT"}‘ or ‘{"decision": "NO_DECISION"}‘. Include any reasoning text in
      the model response. Do not call any other tool.

## Decision Support
Observations may include a DECISION SUPPORT block with precomputed summaries derived from the current book, fees, and limits. These summaries
      are informational only; you remain responsible for the trading decision.

## Objective
Maximise P&L by accepting profitable tenders with the knowledge that the unwind algorithm will work to unwind your order according to the
      behavior described above.



=== USER MESSAGE (tick 24) ===
=== Tick 24 / 300 ===

MARKET STATE (CRZY):
  Best bid: 10.00
  Best ask: 10.03
  Last: 10.03
  ORDER BOOK:
    BIDS ASKS
      10.00 x 5900 10.03 x 500
      10.00 x 4600 10.04 x 4200
       9.99 x 5500 10.05 x 5400
       9.99 x 4600 10.06 x 5100
       9.97 x 4900 10.06 x 5300

MARKET STATE (TAME):
  Best bid: 24.64
  Best ask: 24.67
  Last: 24.69
  ORDER BOOK:
    BIDS ASKS
      24.64 x 4600 24.67 x 2300
      24.52 x 5800 24.67 x 5700
      24.42 x 4500 24.71 x 5400
      24.26 x 4700 24.72 x 4700
      24.18 x 4500 24.74 x 6000

YOUR POSITIONS:
  CRZY: +0 shares (max order: 25,000)
```

Table A5 ***(continued)***

```
  TAME: +0 shares (max order: 10,000)
  Net: +0 shares

YOUR P&L: $0.00
YOUR FINES: $0.00
NLV (net of fines): $0.00

YOUR LIMITS:
  Net: 0 / 100,000
  Gross: 0 / 250,000

TENDER UNDER DECISION:
  Tender ID: 0
  Ticker: TAME
  Side: BUY (your side)
  Quantity: 70,000 shares
  Price: $24.4300
  Expires: in 30 ticks
  Caption: An institution would like to SELL 70000 shares of TAME to you at a price of 24.43. Would you like to BUY the shares from them?

DECISION SUPPORT:
  Tender #0 (TAME):
    Unwind market VWAP (full size): insufficient book depth
    Implied edge / share (net of fee): insufficient book depth
    Implied total edge: insufficient book depth
    Projected position (TAME): +70,000
    Projected net (signed): +70,000
    Projected gross: 70,000
    Would breach net limit: no
    Would breach gross limit: no

Call the ‘tender_decision‘ tool exactly once with ‘{"decision": "ACCEPT"}‘ or ‘{"decision": "NO_DECISION"}‘.
```

Table A6: Less-Directive Prompt Ablation (Normal Speed). Robustness check on the prompt design. The less-directive ablation strips three rush-inducing phrases (*"close positions early"*, *"close all positions before the case ends"*, *"efficiently"*) from the baseline prompt while keeping rules, fines, limits, fees, and tools byte-identical; reruns the normal-speed market with gpt-5.4 across reasoning off / low / medium (50 sessions per prompt × reasoning cell, $N = 300$). Panel A reports cell-level means pooled across both prompts; Δ is less-directive minus baseline within each reasoning level. Panel B reports OLS regressions of performance on reasoning indicators, a prompt indicator, and their interactions, with Wald F-tests on the grouped reasoning, prompt, and interaction coefficients at the bottom. References: reasoning off, prompt baseline. Dollar amounts in thousands ($000s). HC3 robust standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Cell means, baseline vs. less-directive ablation

Prompt	Reasoning	Mean NLV	Std NLV	% Profitable	Mean Trading P&L	Mean Fines	Mean Think Time (ms)	Turns	Tenders Accepted
Baseline	Off	8.3	37.0	62%	22.6	14.3	2,939	75.0	3.7
Less-Directive	Off	−3.2	55.2	58%	13.9	17.1	2,927	75.1	3.9
	Δ	−11.5		−4%	−8.7	2.8	−12	0.1	0.2
Baseline	Low	18.6	46.3	74%	25.5	7.0	3,789	63.3	3.7
Less-Directive	Low	19.0	44.8	72%	29.2	10.2	4,129	59.0	3.9
	Δ	0.5		−2%	3.7	3.2	340	−4.3	0.1
Baseline	Medium	7.3	62.8	62%	28.8	21.4	7,303	40.2	3.3
Less-Directive	Medium	−3.5	60.0	52%	21.8	25.4	8,063	37.9	3.2
	Δ	−10.9		−10%	−6.9	4.0	761	−2.2	−0.1
Baseline	All	11.4	49.8	66%	25.6	14.2	4,677	59.5	3.6
Less-Directive	All	4.1	54.4	61%	21.7	17.6	5,040	57.3	3.7

Panel B: Prompt × Reasoning interaction (pooled $N = 300$)

	(1) NLV	(2) Trading P&L	(3) Fines
Low	10.27	2.94	−7.33*
	(8.47)	(6.60)	(4.11)
Medium	−0.97	6.18	7.15
	(10.41)	(7.63)	(5.65)
Less-Directive	−11.47	−8.66	2.81
	(9.49)	(7.08)	(5.23)
Low × Less-Directive	11.94	12.37	0.43
	(13.22)	(9.97)	(6.57)
Medium × Less-Directive	0.58	1.74	1.15
	(15.62)	(11.47)	(8.11)
Constant	8.31	22.59***	14.28***
	(5.29)	(4.23)	(3.24)
Prompt F	1.46 (0.228)	1.50 (0.222)	0.29 (0.592)
Prompt × Reason F	0.49 (0.616)	0.87 (0.422)	0.01 (0.990)
Reason F	0.86 (0.423)	0.34 (0.711)	4.30 (0.014)
N	300	300	300
R^2	0.0302	0.0181	0.0571

Table A7: Less-Directive Ablation Prompt: Rendered Example. System prompt and user message for the first tender-visible turn of an example less-directive session feeding Table A6 (`level2_ablation.md` as rendered against the live case configuration). The runtime placeholder {`reflection_block`} is empty in this session.

```
=== SYSTEM PROMPT ===
You are a liability trader in a live RIT (Rotman Interactive Trader) case.

## Securities
There are 2 securities: CRZY, TAME.
Each has its own order book. When submitting orders, use the ticker as the "venue" parameter (e.g. venue="CRZY").
Positions are tracked per security.

**Order size limits** (per single order -- RIT will REJECT orders exceeding these):
  CRZY: max 25,000 shares per order
  TAME: max 10,000 shares per order
If you need to trade more than the limit, split into multiple smaller orders.

## Case Structure
Duration: 300 ticks (each tick is ~1 second of real time).
You are called periodically to observe and optionally act. The tick numbers in your observations show how much time has elapsed between your
      turns.

## Your Job: Liability Trading
Institutional tender offers arrive during the case. Each tender specifies a ticker, a side (BUY or SELL from your perspective), a quantity, a
      price, and a deadline.

When a tender is live:
1. Evaluate the tender: check the order book depth for THAT ticker. Can you unwind profitably?
2. If yes, ACCEPT the tender using decide_tender.
3. You may let a tender expire. There is no need to explicitly decline.

After accepting a tender:
1. Unwind the resulting position by submitting orders on the SAME ticker.
2. There is no requirement to trade immediately after acceptance or to act on every observation.

Tenders remain active for approximately 30 ticks (~30 seconds). Due to polling latency, you may not see a tender until a few ticks after it
      is issued. The time_left field shows how many ticks remain before expiration. If you do not accept before the deadline, the tender
      expires automatically.

Do NOT speculate or make markets. Only trade to unwind positions created by accepted tenders.

## Front-Running Prohibition
Do NOT trade a security to anticipate a pending tender. If you have no existing position in a security, you must accept or let expire the
      tender BEFORE placing any orders on that security. Exception: if you are already unwinding a position from a previously accepted
      tender, you may continue closing that position even if a new tender on the same security arrives.

## Position Limits
Net limit: 100,000 shares (sum of all positions)
Gross limit: 250,000 shares (sum of absolute positions)
```

Table A7 ***(continued)***

```
## Transaction Fees
Fee: $0.02 per share on every market trade.
Tender fills do not incur a transaction fee.

## Fines
Front-running or speculative trading (any trade not directly unwinding an accepted tender) incurs fines: $0.20/share up to 5,000 shares,
      $0.40/share beyond that. Uncovered positions at the end of the case are fined at $1.00/share. These fines are assessed in real time and
      reduce your P&L.

**Important:** If you submit unwind orders in the same action as accepting a tender, you may incur front-running fines because the tender has
      not yet settled. Accepting the tender in one step then submitting unwind orders in a subsequent step will avoid these fines.

## End of Case
Aim to be flat (zero inventory on all securities) before the case ends.

## Available Tools
- submit_order: Submit an order. Set venue to the ticker name. Types: LIMIT, MARKETABLE_LIMIT, MARKET.
- cancel_order: Cancel a resting order by order_id.
- decide_tender: Accept or decline a tender by tender_id.
You do not need to act every turn.




## Objective
Maximise P&L by accepting profitable tenders and unwinding positions.




=== USER MESSAGE (turn 5) ===
=== Tick 18 ===

MARKET STATE (CRZY):
  Best bid: 10.13
  Best ask: 10.15
  Last: 10.12
  Volume: 81,800 My volume: 81,800
  ORDER BOOK:
    BIDS ASKS
       10.13 x 4200 10.15 x 5500
       10.09 x 4300 10.16 x 4300
       10.08 x 1300 10.17 x 5700
       10.06 x 5100 10.18 x 5000
       10.04 x 1200 10.21 x 4800
       10.03 x 5300 10.24 x 4900
       10.00 x 2000 10.26 x 5100
       10.00 x 5600 10.27 x 4200
        9.98 x 4900 10.58 x 4200
        9.97 x 3300
        9.97 x 4600
        9.96 x 5700
        9.95 x 5800
        9.94 x 4100
```

Table A7 ***(continued)***

```
       9.94 x 5500
       9.88 x 5200
       9.86 x 4400
       9.86 x 4400
       9.85 x 5800
       9.84 x 4500

MARKET STATE (TAME):
  Best bid: 24.70
  Best ask: 24.71
  Last: 24.70
  Volume: 95,700 My volume: 95,700
  ORDER BOOK:
    BIDS ASKS
      24.70 x 4700 24.71 x 4800
      24.69 x 4900 24.77 x 600
      24.69 x 4700 24.80 x 900
      24.69 x 4500 24.82 x 5800
      24.68 x 4700 24.83 x 2300
      24.67 x 5900 24.83 x 4300
      23.96 x 4900 24.84 x 6000
                           24.85 x 4700
                           24.87 x 5400
                           24.88 x 4200
                           24.90 x 4300
                           24.94 x 5600
                           24.96 x 4700
                           24.98 x 5000
                           24.99 x 5500
                           25.03 x 1300
                           25.03 x 4300
                           25.03 x 5900
                           25.07 x 2900
                           25.09 x 4300

YOUR POSITIONS:
  CRZY: 0 shares (max order: 25000) P&L: +0.00
  TAME: 0 shares (max order: 10000) P&L: +0.00
  Net: 0 shares

YOUR P&L: $0.00
YOUR FINES: $0.00
NLV (net of fines): $0.00

YOUR LIMITS:
  Net: 0 / 100,000 (100,000 remaining)
  Gross: 0 / 250,000 (250,000 remaining)

TENDERS:
  [0] "An institution would like to BUY 74000 shares of TAME from you at a price of 25.01. Would you like to SELL the shares to them?"
        (expires in 27 ticks)
```

Table A8: Reasoning, Response Time, and Performance (OpenAI Model Scale). Extension of Table 2 to the three-model OpenAI sample (gpt-5.4, gpt-5.4-mini, gpt-5.4-nano; 50 sessions per model × reasoning cell; $N = 450$). OLS regressions of NLV (cols 1–2), Trading P&L (cols 3–4), and Fines (cols 5–6) on model indicators, reasoning indicators, and their interaction. Columns (2), (4), and (6) add log average think time per agent turn. NLV equals Trading P&L minus Fines. Dollar amounts in thousands ($000s). HC3 robust standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels. $N = 450$ (50 sessions per cell). Reference: gpt-5.4, reasoning off.

	(1) NLV	(2) NLV	(3) TP&L	(4) TP&L	(5) Fines	(6) Fines
gpt-5.4-mini	−35.16***	−62.19***	−12.76**	−28.72***	22.40***	33.47***
	(9.42)	(9.72)	(6.40)	(6.56)	(6.77)	(7.03)
gpt-5.4-nano	−6.88	−32.20***	−16.99***	−31.94***	−10.11**	0.26
	(5.99)	(6.62)	(5.00)	(5.38)	(4.08)	(4.38)
Low	10.27	31.22***	2.94	15.30**	−7.33*	−15.91***
	(8.47)	(8.23)	(6.60)	(6.35)	(4.11)	(4.49)
Medium	−0.97	74.26***	6.18	50.60***	7.15	−23.66***
	(10.41)	(10.68)	(7.63)	(8.51)	(5.65)	(6.76)
gpt-5.4-mini × Low	−3.08	12.13	−4.05	4.94	−0.96	−7.20
	(13.23)	(12.74)	(9.65)	(9.43)	(9.28)	(9.13)
gpt-5.4-mini × Medium	33.29**	65.17***	8.06	26.89***	−25.23***	−38.29***
	(15.57)	(14.31)	(11.08)	(10.39)	(9.26)	(9.06)
gpt-5.4-nano × Low	−31.62***	−25.31**	1.45	5.17	33.06***	30.48***
	(11.34)	(10.74)	(8.18)	(7.96)	(6.97)	(6.74)
gpt-5.4-nano × Medium	0.66	−9.52	7.36	1.35	6.70	10.87
	(12.79)	(11.26)	(9.61)	(8.67)	(7.29)	(7.09)
ln(Think Time)		−88.69***		−52.36***		36.33***
		(9.24)		(7.90)		(5.18)
Constant	8.31	−11.10**	22.59***	11.14**	14.28***	22.23***
	(5.29)	(5.56)	(4.23)	(4.49)	(3.24)	(3.55)
N	450	450	450	450	450	450
R^2	0.0854	0.2478	0.0522	0.1649	0.0986	0.1659
Model × Reasoning	$F(4, 441) = 3.78$***	$F(4, 440) = 10.16$***	$F(4, 441) = 0.36$	$F(4, 440) = 2.47$**	$F(4, 441) = 9.17$***	$F(4, 440) = 13.12$***
	$p = 0.005$	$p < 0.001$	$p = 0.838$	$p = 0.044$	$p < 0.001$	$p < 0.001$

Table A9: Reasoning, Response Time, and Performance (Anthropic Family). Anthropic analog of Table 2: claude-opus-4-6 and claude-sonnet-4-6 × off/low/medium extended thinking, 50 sessions per cell ($N = 300$). OLS regressions of NLV (cols 1–2), Trading P&L (cols 3–4), and Fines (cols 5–6) on model indicators, reasoning indicators, and their interaction. Columns (2), (4), and (6) add log average think time per agent turn. References: Opus 4.6, reasoning off. Dollar amounts in thousands ($000s). HC3 robust standard errors in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	(1) NLV	(2) NLV	(3) Trading P&L	(4) Trading P&L	(5) Fines	(6) Fines
Sonnet 4.6	2.48	13.47**	6.01	15.07***	3.53**	1.61
	(6.18)	(6.45)	(5.74)	(5.84)	(1.58)	(2.04)
Low	−3.59	4.05	−2.31	4.00	1.28	−0.05
	(5.75)	(5.59)	(5.31)	(5.11)	(1.73)	(1.95)
Medium	−23.71***	10.50	−12.80	15.43*	10.91***	4.93
	(8.58)	(10.06)	(7.79)	(8.08)	(3.66)	(5.47)
Sonnet 4.6 × Low	−6.52	25.60**	1.54	28.05***	8.06*	2.45
	(8.92)	(11.41)	(8.24)	(10.27)	(4.14)	(5.05)
Sonnet 4.6 × Medium	3.05	47.28***	−3.33	33.17**	−6.38	−14.12**
	(11.42)	(14.51)	(9.97)	(13.41)	(4.85)	(6.12)
ln(Think Time)		−55.95***		−46.17***		9.78
		(11.27)		(9.54)		(6.23)
Constant	28.41***	−3.76	29.34***	2.80	0.94	6.56*
	(3.60)	(7.31)	(3.38)	(6.36)	(0.66)	(3.65)
N	300	300	300	300	300	300
R^2	0.0617	0.1709	0.0448	0.1416	0.0709	0.0858
Model × Reasoning	$F(2, 294) = 0.44$	$F(2, 293) = 5.32$***	$F(2, 294) = 0.12$	$F(2, 293) = 4.11$**	$F(2, 294) = 3.13$**	$F(2, 293) = 4.03$**
	$p = 0.646$	$p = 0.005$	$p = 0.889$	$p = 0.017$	$p = 0.045$	$p = 0.019$

Table A10: Case Parameters (ALGO3). Case parameters for the ALGO3 dual-venue experiment. A single underlying security is traded on two venues (THOR_M, THOR_A); position and gross limits aggregate across both books. All sessions use the same case configuration with randomized price paths.

Parameter	Value
Market structure	
Underlying	Single stock traded on two venues (THOR_M, THOR_A)
Order books	Independent per venue; shares fungible across venues
Duration	300 ticks ($\approx$ 300 seconds)
Periods	1
Position limits (aggregated across both venues)	
Net limit	100,000 shares
Gross limit	200,000 shares
Tender offers	
Arrival	Institutional tenders arrive throughout the session
Duration	$\approx$ 30 ticks per tender
Transaction fees and rebates	
Venue fees	Venue-dependent (charged on every non-tender fill)
Passive-limit rebate	\$0.0075/share on LIMIT fills on the alternative venue
Marketable limits	Treated as market orders; no rebate
Tender fills	Free (no transaction fee)
Penalties	
Front-running fine	\$0.20/share $\leq$ 5,000 shares; \$0.40/share beyond
End-of-case fine	\$1.00/share on uncovered positions
Experimental design	
Model	gpt-5.4
Reasoning levels	Off, Low, Medium
Sessions per cell	50
Total sessions	150 (3 cells $\times$ 50)
Agent polling	$\approx$ 1 second between observations

Table A11: Trading Performance with Reflection Between Sessions. Mean session NLV in $000s; 50 three-session chains per configuration. Session 1 has no playbook. Before sessions 2 and 3, a separate `gpt-5.4` model with `xhigh` reasoning reviews the preceding session and revises a playbook of up to ten rules. Price paths and tender schedules are independently redrawn each session; model parameters remain fixed. Session 3–1 reports the mean within-chain change, with its standard error below in parentheses. The pooled estimate averages all 600 chains. *** denotes significance at the 1% level. Changes are calculated before rounding.

Market speed	Reasoning	Session 1	Session 2	Session 3	Session 3–1
		Panel A: `gpt-5.4`			
Slow	Off	11.9	11.1	19.7	7.8
					(7.0)
	Low	20.8	27.2	31.5	10.7
					(6.4)
	Medium	20.6	20.6	27.8	7.2
					(6.3)
Normal	Off	13.7	17.1	24.2	10.5
					(6.6)
	Low	24.3	27.8	25.6	1.4
					(5.8)
	Medium	23.8	20.3	19.7	−4.1
					(9.1)
Fast	Off	5.2	11.0	15.2	10.0
					(8.7)
	Low	21.8	10.9	14.5	−7.2
					(7.4)
	Medium	4.2	12.9	8.6	4.4
					(10.3)
		Panel B: `gpt-5.4-mini`			
Normal	Off	−53.5	−14.1	−30.2	23.2
					(14.0)
	Low	−50.2	−0.9	20.9	71.1***
					(13.1)
	Medium	11.0	11.3	19.3	8.3
					(10.9)
Mean change across all configurations					11.9***
					(2.7)

Table A12: ALGO3 Prompt: Rendered Example. System prompt and user message for the first tender-visible turn of an example ALGO3 session feeding Table A13.

```
=== SYSTEM PROMPT ===
You are a liability trader in a live RIT (Rotman Interactive Trader) case.

## Securities
The case has a single underlying stock traded on TWO venues: THOR_M, THOR_A. Each venue has its own order book, but both books trade the SAME
      underlying security. A share bought on one venue is fungible with a share sold on the other.

When submitting orders, set the "venue" parameter to the specific ticker you want the order routed to (e.g. venue="THOR_M").

The RIT API reports the SAME underlying position on each venue's ticker -- the per-venue rows in `YOUR POSITIONS` mirror each other. **Do NOT
      sum them.** The `Net:` line is your single underlying position; `YOUR P&L`, `YOUR FINES`, and `YOUR LIMITS` likewise report on the
      underlying as a whole, not per venue.

**Order size limits** (per single order -- RIT will REJECT orders exceeding these):
  THOR_M: max 25,000 shares per order
  THOR_A: max 25,000 shares per order
If you need to trade more than the limit, split into multiple smaller orders.

## Case Structure
Duration: 300 ticks (each tick is ~1 second of real time).
You are called periodically to observe and optionally act. The tick numbers in your observations show how much time has elapsed between your
      turns.

## Your Job: Liability Trading with Two Venues
Institutional tender offers arrive during the case. Each tender specifies a ticker, a side (BUY or SELL from your perspective), a quantity, a
      price, and a deadline.

Your job:
1. Evaluate the tender. Consider the book depth on BOTH venues for the underlying -- the liquidity you can unwind against lives across the
      two books combined, not just the venue the tender is quoted on.
2. If the tender is profitable after fees, ACCEPT it using decide_tender.
3. Unwind the acquired position by submitting orders on either or both venues.
4. Tenders remain active for approximately 30 ticks (~30 seconds). Due to polling latency, you may not see a tender until a few ticks after
      it is issued. The time_left field shows how many ticks remain before expiration. If you do not accept before the deadline, the tender
      expires automatically -- there is no need to explicitly decline.

Do NOT speculate or make markets. Only trade to unwind accepted tenders.

## Front-Running Prohibition
Do NOT trade a security to anticipate a pending tender. If you have no existing position in the underlying (zero on BOTH venues), you must
      accept or let the tender expire BEFORE placing any orders. Exception: if you are already unwinding a position from a previously
      accepted tender, you may continue closing that position even if a new tender arrives. You may unwind on either venue regardless of
      which venue the tender settled on -- they are the same stock.

## Position Limits
Net limit: 100,000 shares
Gross limit: 200,000 shares
Both limits apply to the single underlying position (see Securities above on how mirrored rows work).
```

Table A11 ***(continued)***

```
## Transaction Fees and Rebates
Fees per share on every market trade:
  THOR_M: $0.01
  THOR_A: $0.005
The two venues charge different fees. In addition, passive LIMIT orders filled on the alternative venue (THOR_A) earn a rebate of $0.0075 per
      share net. Marketable limit orders (limit orders that cross the spread) are treated as market orders and do NOT earn the rebate.
Tender fills do not incur a transaction fee.

## Fines
Front-running or speculative trading (any trade not directly unwinding an accepted tender) incurs fines: $0.20/share up to 5,000 shares,
      $0.40/share beyond that.

Uncovered positions at the end of the case are fined at $1.00/share.

**Important:** If you submit unwind orders in the same action as accepting a tender, you may incur front-running fines because the tender has
      not yet settled. Accepting the tender in one step then submitting unwind orders in a subsequent step will avoid these fines.

## End of Case
Aim to be flat (zero inventory on BOTH venues) before the case ends. Monitor the tick count and close positions early.

## Available Tools
- submit_order: Submit an order. Set venue to the specific ticker (e.g. "THOR_M"). Types: LIMIT, MARKETABLE_LIMIT, MARKET.
- cancel_order: Cancel a resting order by order_id.
- decide_tender: Accept or decline a tender by tender_id.
You do not need to act every turn.



## Objective
Maximise P&L by accepting profitable tenders and unwinding positions efficiently across the two venues. Manage risk and close all positions
      on BOTH venues before the case ends.




=== USER MESSAGE (turn 21) ===
=== Tick 57 ===

MARKET STATE (THOR_M):
  Best bid: 10.09
  Best ask: 10.10
  Last: 10.10
  Volume: 84,400 My volume: 84,400
  ORDER BOOK:
    BIDS ASKS
      10.09 x 3600 10.10 x 1700
      10.09 x 2800 10.16 x 2700
      10.09 x 2700 10.16 x 3000
      10.09 x 2800 10.17 x 3500
      10.08 x 2500 10.18 x 200
      10.07 x 2700 10.19 x 2000
      10.07 x 2800 10.19 x 2600
```

Table A11 ***(continued)***

```
      10.02 x 1900 10.22 x 3200
      10.01 x 2800 10.22 x 3200
      10.01 x 3600 10.25 x 2600
      10.01 x 2800 10.26 x 2500
       9.99 x 2400 10.27 x 3200
       9.99 x 2700 10.29 x 2500
       9.98 x 3500 10.33 x 2900
       9.98 x 3100 10.45 x 2900
       9.98 x 2700 11.03 x 2600
       9.97 x 2700
       9.97 x 2800
       9.96 x 3600
       9.95 x 100

MARKET STATE (THOR_A):
  Best bid: 10.10
  Best ask: 10.17
  Last: 10.17
  Volume: 88,800 My volume: 88,800
  ORDER BOOK:
    BIDS ASKS
      10.10 x 5200 10.17 x 4300
      10.09 x 6200 10.20 x 500
      10.09 x 6100 10.20 x 6100
      10.09 x 5100 10.21 x 800
      10.07 x 4900 10.21 x 5400
       9.99 x 800 10.22 x 5900
       9.96 x 5400 10.22 x 5100
       9.96 x 5900 10.24 x 6100
       9.96 x 5200 10.28 x 5700
       9.95 x 6300 10.28 x 6400
       9.59 x 6400 10.29 x 5500
       9.58 x 6200 10.32 x 5500
       9.48 x 5600 10.39 x 4900
       9.15 x 6400
       9.14 x 6000
       9.14 x 5400

YOUR POSITIONS:
  THOR_M: 0 shares (max order: 25000) P&L: +0.00
  THOR_A: 0 shares (max order: 25000) P&L: +0.00
  Net: 0 shares

YOUR P&L: $0.00
YOUR FINES: $0.00
NLV (net of fines): $0.00

YOUR LIMITS:
  Net: 0 / 100,000 (100,000 remaining)
  Gross: 0 / 200,000 (200,000 remaining)

TENDERS:
  [0] "An institution would like to SELL 72000 shares of THOR_M to you at a price of $10.11. Would you like to BUY the shares from them?"
```

Table A11 *(continued)*

```
        (expires in 29 ticks)
```

Table A13: ALGO3 Results. ALGO3 extends LT3 to a single underlying traded on two venues (THOR_M, THOR_A) with venue-dependent fees and a passive-limit rebate; position and gross limits aggregate across both books. Panel A reports descriptive means for the deterministic ALGO3 algorithm benchmark and for gpt-5.4 across the three reasoning levels (50 sessions per cell, same columns as Table 9 Panel A). Panel B compares the ALGO3 "gpt-5.4 / Low" cell from Panel A to the algorithm benchmark. The "gpt-5.4 / Low" coefficient is the mean difference between the LLM cell and algorithm runs. Columns: (1) NLV OLS, (2) NLV IQR ($q_{75} - q_{25}$), (3) Trading P&L OLS, (4) Fines OLS. Dollar amounts in thousands ($000s). HC3 robust standard errors for OLS in parentheses; bootstrap SEs for IQR. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Descriptive means

Cell	Mean NLV	Std NLV	Sharpe	% Profitable	Mean Trading P&L	Mean Fines	Mean Think Time (ms)	Tenders Accepted
ALGO3 Algo	48.5	23.3	2.08	100%	49.3	0.9	46	2.7
gpt-5.4 / Off	15.8	29.2	0.54	86%	22.8	6.9	2,334	3.5
gpt-5.4 / Low	18.1	38.0	0.48	74%	25.7	7.6	4,197	3.5
gpt-5.4 / Medium	−40.9	82.5	-0.50	50%	21.1	62.0	9,040	2.4

Panel B: LLM-vs-algorithm regressions

	(1) NLV OLS	(2) NLV IQR	(3) TP&L OLS	(4) Fines OLS
gpt-5.4 (Low)	−30.33***	7.37	−23.61***	6.72*
	(6.37)	(10.27)	(5.04)	(4.03)
Constant	48.45***	39.59***	49.32***	0.86
	(3.33)	(6.95)	(3.22)	(0.61)
N	100	100	100	100
R^2	0.1910		0.1858	0.0281

Table A14: Network Size and NN Trader Performance. Mean session NLV in $000s; 50 sessions per architecture and speed. One network is varied while the other remains at its baseline specification. Inputs and fitted parameters are held fixed across speeds. † marks the baseline specifications, which represent the same trader. Panel B jointly tests whether the effect of market speed on mean NLV is the same across architectures, using HC3-robust standard errors. Neither test rejects this null at the 10% level.

Panel A: Performance by network size

Hidden layers × units	Parameters	Mean session NLV Normal	Slow	Fast
Selection network varied				
$1 \times 8^{\dagger}$	49	42.1	38.5	40.6
1×32	193	37.9	44.4	40.7
1×128	769	43.1	43.8	46.3
2×8	121	43.9	40.7	44.0
2×32	1,249	42.1	42.2	46.0
2×128	17,281	38.2	36.1	40.4
3×8	193	37.3	39.7	42.9
3×32	2,305	44.7	40.2	43.0
3×128	33,793	43.0	43.3	42.4
Mean across architectures		41.4	41.0	42.9
Execution network varied				
1×32	885	39.7	35.4	37.7
1×128	3,477	36.5	33.0	37.5
1×512	13,845	40.6	38.2	34.4
2×32	1,941	41.3	40.6	41.8
$2 \times 128^{\dagger}$	19,989	42.1	38.5	40.6
2×512	276,501	39.0	42.4	41.9
3×32	2,997	41.0	39.1	37.4
3×128	36,501	37.4	41.8	35.1
3×512	539,157	39.3	42.4	36.7
Mean across architectures		39.7	39.0	38.1

Panel B: Architecture-by-speed interaction tests

Network varied	HC3-robust F
Selection	0.50
Execution	0.59

Online Appendix: A Second Task

Our final robustness exercise asks whether the main patterns are specific to LT3 or extend to a related but more complex execution task. The baseline LT3 environment has two securities, each traded in a single order book. In many electronic markets, however, execution also requires routing across venues with different depth, fees, and rebates. This setting is useful for our question because it increases the value of real-time information processing while preserving the same economic structure: the trader must decide whether to accept a client-style tender and then unwind the resulting inventory without speculating.

We study this extension using the ALGO3 case. In ALGO3, a single underlying security, THOR, trades on two venues with independent order books, `THOR_A` and `THOR_M`. Shares are fungible across venues, but fees and rebates differ, and position limits apply to the aggregate exposure. The agent must therefore evaluate tender profitability using the combined liquidity across both books and route unwind orders across venues. This is a natural second task for the paper because it adds fragmentation and routing complexity without changing the central principal-liquidity-provision problem. Online Appendix Tables A10 and A12 report the ALGO3 case parameters and the corresponding two-venue playbook prompt.

The ALGO3 algorithm we consider is a simple strategy that accepts only tenders arriving on `THOR_M` whose net-of-fee edge is at least 5 cents. It then unwinds by routing market orders of up to 5,000 shares, clipped to the chosen venue's top-of-book size, once per tick to whichever of `THOR_M` ($0.01/share fee) or `THOR_A` ($0.005/share fee) offers the better top quote, until the tender quantity, typically 70,000–90,000 shares, is fully unwound.

Online Appendix Table A13 reports the results. Panel A shows that the agent can operate in the fragmented environment, but the same non-monotone reasoning pattern appears. With reasoning off, `gpt-5.4` earns mean NLV of $15.8 thousand and is profitable in 86% of sessions. Low reasoning raises mean NLV slightly to $18.1 thousand, but with higher dispersion and a lower profitable-session rate of 74%. Medium reasoning performs substantially worse: mean NLV falls to −$40.9 thousand and average fines rise to $62.0 thousand. As in LT3, larger reasoning budgets do not mechanically improve trading performance, and the highest reasoning setting creates substantial execution risk.

Table A15: Trading Performance Under Decision Delay. Mean session NLV in $000s; 50 sessions per cell. NN-baseline uses networks trained without delay; NN-retrained uses networks retrained under the imposed delays. Off, Low, and Medium denote the LLM agent's reasoning settings and, for the NN trader, delays drawn from the corresponding response-time distributions. Reported ticks are median delays when a tender is live. Medium–Off is the change in mean NLV from Off to Medium; Panel D reports the fast-market change minus the slow-market change. Standard errors appear in parentheses below estimates. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively, using Welch tests.

Trader	No delay	Off	Low	Medium	Medium–Off
Panel A: Fast market (median delays 5, 11, and 19 ticks)					
LLM agent	—	10.7	14.0	−18.4	−29.1**
					(11.3)
NN-unaware	45.0	23.1	1.0	−11.2	−34.3***
					(8.2)
NN-aware	—	32.1	22.9	13.7	−18.4***
					(4.8)
Panel B: Normal market (median delays 3, 8, and 13 ticks)					
LLM agent	—	8.3	18.6	7.3	−1.0
					(10.3)
NN-unaware	37.5	35.7	7.1	−17.1	−52.8***
					(7.8)
NN-aware	—	39.3	38.3	16.1	−23.1***
					(4.3)
Panel C: Slow market (median delays 2, 4, and 7 ticks)					
LLM agent	—	2.7	22.9	27.8	+25.2***
					(8.2)
NN-unaware	42.1	43.2	40.6	−0.9	−44.1***
					(7.7)
NN-aware	—	47.5	33.1	28.8	−18.7***
					(4.7)

Panel D: Difference in the Medium–Off change, fast minus slow

Trader	Difference	p-value
LLM agent	−54.3***	< 0.001
	(14.0)	
NN-unaware	+9.7	0.389
	(11.3)	
NN-aware	+0.3	0.968
	(6.7)	

Panel A also compares the agent with a deterministic ALGO3 benchmark algorithm. The algorithm earns mean NLV of $48.5 thousand, compared with $18.1 thousand for the low-reasoning agent. It also earns higher Trading P&L while paying lower average fines: $49.3 thousand in Trading P&L and $0.9 thousand in fines for the algorithm, compared with $25.7 thousand and $7.6 thousand for the low-reasoning agent. The algorithm accepts fewer tenders on average, 2.7 compared with 3.5 for the low-reasoning agent, but converts those opportunities into higher and cleaner performance.

Panel B reports the comparison formally. The low-reasoning agent's NLV is $30.3 thousand below the algorithm. The gap reflects both lower Trading P&L (−$23.6 thousand) and higher fines (+$6.7 thousand). The interquartile-range difference is not statistically precise, so the evidence is strongest for a mean-performance gap rather than a distributional one. The fragmented-execution benchmark therefore reinforces the compiled-policy interpretation from LT3: when the routing, tender-evaluation, and unwind logic can be specified in advance, the algorithm avoids much of the real-time deliberation cost faced by the agent.

The fragmented-execution results reinforce the scope of the paper's main argument. The agent remains viable under off and low reasoning, but additional reasoning is not uniformly valuable. Medium reasoning again underperforms, and the deterministic benchmark remains strong because it avoids real-time deliberation and operational execution errors. The evidence shows that the speed–reasoning frontier is not an artifact of the LT3 design.

Online Appendix: Neural Network Trader

This appendix describes the training procedures for the NN trader. The execution network is fitted by value iteration, while the selection network is fitted using average session NLV with the execution policy held fixed. Both networks are trained in an offline simulator of LT3. Table **??** summarizes their architecture and training settings. An execution episode begins with a position of X shares, long or short with equal probability. Training fixes $X = 71{,}000$. Let P_0 be the bid-ask midpoint when the position arrives, and let $\varsigma = 1$ for a long position and $\varsigma = -1$ for a short position. A long position is unwound by selling and a short position by buying. During execution training, the trader makes one decision per tick over a pacing window of H decisions. If only L decision ticks remain before the deadline, the episode contains $h = \min\{H, L\}$ decisions. The decisions occur at $t = 0, \ldots, h-1$. Any inventory remaining at $t = h$ incurs the terminal penalty.

Let x_t denote inventory remaining before decision t, with $x_0 = X$. The network observes

$$s_t = \Big(\tau_t/H,\ x_t/X,\ \tilde{m}_t,\ a_{t-1}\Big), \qquad \tau_t = h - t, \qquad \tilde{m}_t = (m_t - 100)/100. \tag{9}$$

Here m_t is the last-traded price and a_{t-1} is the previous liquidation fraction, initialized at zero. The action set $\mathcal{G}$ contains 21 liquidation fractions: zero and 20 positive fractions generated geometrically from 2% to 100%. Action a_t submits approximately $a_t x_t$ shares as market orders in the unwinding direction, rounded to a whole number of shares and split into orders within the maximum order size. If n_t shares fill, inventory evolves as

$$x_{t+1} = x_t - n_t. \tag{10}$$

When orders fill in full, this is $x_{t+1} = (1 - a_t)x_t$ up to rounding. Unfilled market-order quantities are cancelled. For fills of n_f shares at prices p_f, implementation shortfall relative to the arrival midpoint is

$$\mathrm{IS}_t = \sum_f \big[\varsigma(P_0 - p_f) + \phi\big] n_f, \qquad \phi = \$0.02. \tag{11}$$

The decision reward is $r_t = -\mathrm{IS}_t/X$. Inventory remaining when the pacing window closes or the session deadline arrives, whichever comes first, incurs a terminal penalty of $\Phi = \$50$ per share. The execution objective is

$$\max_{\pi} \ \mathbb{E}\left[\sum_{t=0}^{h-1} r_t - \Phi\frac{x_h}{X}\right], \tag{12}$$

where π maps the network's state to a liquidation fraction. The terminal penalty is a training incentive to complete liquidation.

Fitting the Execution Network The network $Q_w(s,a)$ returns an estimated value for each of the 21 actions. Parameters of the network are initialized randomly and fitted by fitted value iteration, a reinforcement-learning method. For each sampled market state, we simulate every liquidation fraction under independently redrawn order flow. The resulting rewards and next states are used in fitted value iteration. Let $\widehat{r}_{iak}$ denote this reward for state i, action a, and draw k, including the terminal inventory penalty when the episode ends. Let $d_{iak} = 1$ on a terminal transition and zero otherwise. With target-network parameters w^{-}, the regression target is

$$y_i(a) = \frac{1}{5}\sum_{k=1}^{5}\left[\widehat{r}_{iak} + (1-d_{iak})\,\mathrm{clip}\left(\max_{a'\in\mathcal{G}} Q_{w^{-}}(s'_{iak}, a'), -3, 0\right)\right]. \tag{13}$$

The clipping bounds the continuation value between -3 and zero to limit the influence of large predicted future values on the training target. For a minibatch $\mathcal{B}$ of sampled states, the fitting objective is

$$\min_{w} \ \frac{1}{21|\mathcal{B}|}\sum_{i\in\mathcal{B}}\sum_{a\in\mathcal{G}} \ell\big(Q_w(s_i,a) - y_i(a)\big), \tag{14}$$

where ℓ is the Huber loss, quadratic for residuals with absolute value below one and linear outside that range. We optimize the network with Adam at a learning rate of 10^{-3}. Each of 120 fitted value-iteration rounds comprises 150 gradient updates, using minibatches of 256 states sampled with replacement. We recompute the targets at the start of each round, then

update the target-network parameters at its end according to $w^- \leftarrow 0.9w^- + 0.1w$. The fitted policy chooses the action with the highest predicted value.

Fitting the Selection Network We initialize the network to approximate a 15-cent book-edge threshold, with parameters w_0. Holding the execution policy ψ fixed, we score candidate parameters w using

$$J(w) = \frac{1}{24}\sum_{n=1}^{24} \mathrm{NLV}^{(n)}(w;\psi) - \lambda_w \|w - w_0\|_2, \tag{15}$$

where NLV is measured in dollars, and λ_w denotes the penalty coefficient that discourages large departures from the initial weights. We use the cross-entropy method (CEM) to search over the network's parameters. At each iteration, CEM draws candidate parameter vectors from a normal distribution, evaluates each candidate's average NLV on training sessions, and shifts the distribution toward the best-performing candidates. We then evaluate the selected policy out of sample on 60 held-out test sessions that were not used in the parameter search.